RETC Application: Scientific Background and User Guide

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1. Introduction
2. Workflow overview
3. Problem definition, data input and units
4. Retention models
5. Parameter estimation
6. Uncertainty: 95% confidence intervals
7. Specific points: field capacity, wilting point and plant-available water
8. Pore-size distribution and pore classes
9. Hydraulic conductivity predicted from the retention curve
10. Outputs
11. Software design and verification
12. Assumptions and limits of use
13. References

1. Introduction

The RETC Application fits soil water retention curves to measured data, reports every derived quantity with a 95% confidence interval, and predicts hydraulic conductivity and pore-size distributions from the fitted curve. It was developed at the Chair of Soil Biophysics and Environmental Systems, Technical University of Munich, for pressure plate data, and is built so that the same core can serve evaporation (HYPROP) data.

The soil water retention curve, θ(h), links the volumetric water content θ to the matric potential h. It controls how much water a soil stores, how much of it plants can extract, and, through the pore system it reflects, how fast water moves. Measured retention data are discrete and noisy. A parametric model fitted to them turns the points into a continuous function that can be interpolated, compared between treatments, and passed to flow models.

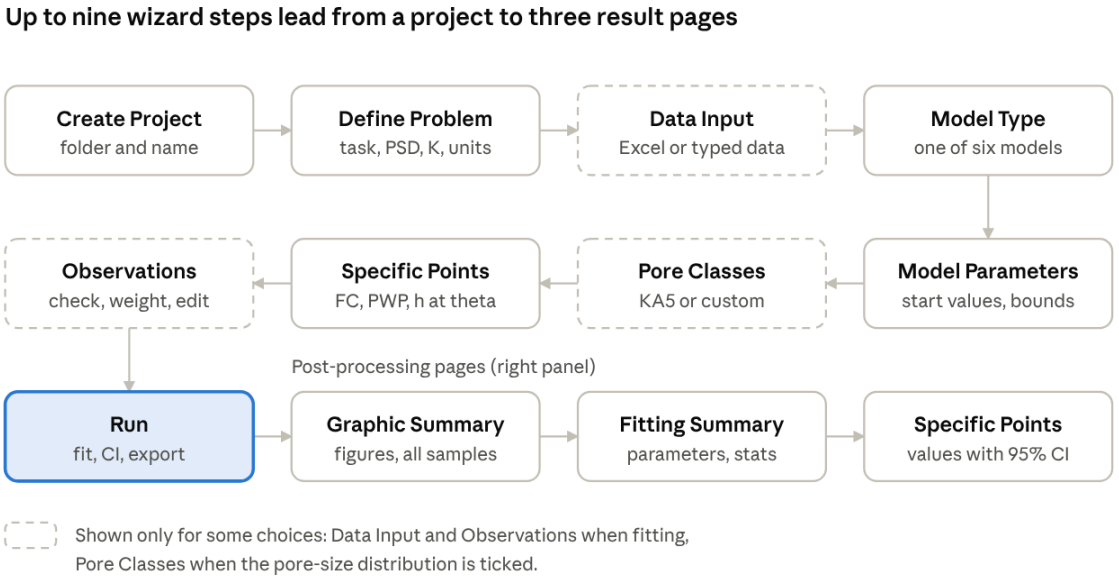
The name follows RETC (van Genuchten et al., 1991), the reference program for this task. The app keeps that purpose and adds what current practice needs:

- **Six retention models**: van Genuchten, Brooks–Corey, Kosugi, bimodal van Genuchten (Durner), and the PDI variants of the unimodal and bimodal van Genuchten models that describe adsorbed water in the dry range.
- **Uncertainty on every result**: parameters, the fitted curve, water content at given heads, plant-available water, matric potential at given water contents, pore-class volumes and conductivity, each with a two-sided 95% confidence interval.
- **Derived soil properties**: field capacity, permanent wilting point and plant-available water content; the pore-size distribution and pore volumes in diameter classes; the unsaturated hydraulic conductivity K(h), with a measured or a predicted saturated conductivity.
- **Free choice of units**: matric potential in cm, hPa or kPa; water content in cm^3/cm^3 or %; conductivity in six common units. Inputs, parameters, figures and files all use the units chosen.
- **Batch processing of treatments**: each sample in a data set is fitted separately, with replicates entering one least-squares fit, and all samples are compared in shared figures and one summary workbook.
- **Transparent, tested code**: a small set of Python modules, each with one task, checked against analytical solutions, numerical integration, published reference formulas and Monte Carlo coverage tests.

This document explains the scientific basis of each part of the app and how to use it. Section 4 presents the six retention models with their original references. Sections 5 to 9 describe the fitting, the uncertainty analysis and the derived properties. Sections 10 to 12 cover outputs, software design and limits of use.

2. Workflow overview

The app is a wizard. A list on the left shows the pre-processing steps reached so far; a list on the right opens the result pages after a run. Only the steps that apply to the current choices are shown, in a fixed order, so the path through the app always matches the problem defined.



The app solves two tasks, chosen on the Define Problem page:

- **Fit retention curve based on observation.** Measured (h, θ) pairs are imported or typed in. The selected model is fitted to each sample by weighted least squares, and all results carry 95% confidence intervals.
- **Estimate retention curve based on known model variables.** No data are needed. The user enters parameter values (for example from a database or a pedotransfer function), and the app computes the curve, the specific points, the pore-size distribution and K(h) from them. Confidence intervals are not defined in this mode, because the parameters carry no estimated covariance.

The steps, in order:

1. **Create Project.** Name the project and choose its folder. All outputs are written there, and the project file stores every setting so a session can be reopened.
2. **Define Problem.** Choose the task; tick the pore-size distribution (PSD) and hydraulic conductivity options; set the units of h, θ and, if K is requested, K. Short hints on the page explain where each choice is used later.
3. **Data Input** (fitting only). Import an Excel workbook (one sheet or column block per sample) or type the data in a table.
4. **Model Type.** Select one of the six retention models. The page shows the model equation and its parameters.
5. **Model Parameters.** Set the start value, lower and upper bound, and fitted or fixed status of each parameter, in the user's units. If K is requested, the K parameters are set here too (Ksat or τ_s, τ, h_clip, vapor flow).
6. **Pore Classes** (PSD only). Use the KA5 default classes or define custom diameter classes.
7. **Define Specific Points.** Set the matric potentials for field capacity and permanent wilting point, plus any extra heads or water contents of interest.
8. **Observations** (fitting only). Review the imported data, assign weights, and remove or correct points. Decimal commas and unit changes are handled here.
9. **Run.** Fit or evaluate every sample, compute the confidence intervals, and write all output files.

After the run, three pages present the results: **Graphic Summary** (all figures, with a properties page for colors, markers, labels and axis limits), **Fitting Summary** (parameters, standard errors, confidence intervals, correlations and goodness of fit) and **Specific Points** (derived quantities with confidence intervals). Section 10 describes them. Every page also has a Help button with short instructions, and the Help tab in the menu bar opens this guide as a PDF.

3. Problem definition, data input and units

3.1 Define Problem

The Define Problem page sets everything that changes the path through the wizard. Its four blocks follow the order in which the choices are used: task, pore-size distribution, hydraulic conductivity and units. Each block carries a short hint on where the choice is used later.

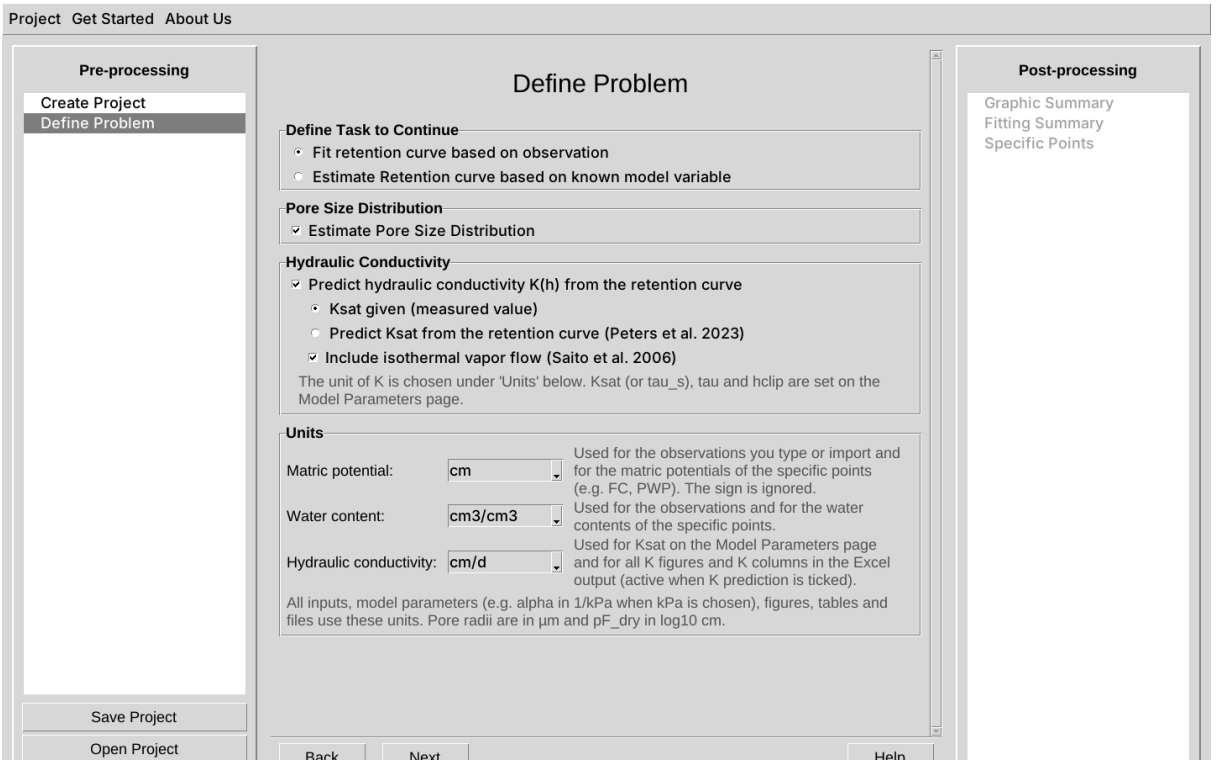


Figure 3.1. Define Problem page. Here the fitting task is chosen, the pore-size distribution is ticked, and K(h) is predicted from the retention curve with a measured Ksat and isothermal vapor flow.

3.2 Units

The app computes in centimeters of water head, volume fractions and cm/d. The user may enter and read everything in other common units. The choice is made once, on the Define Problem page, and applies everywhere: typed or imported data, start values and bounds, specific points, figures, tables and output files. Results are converted once, after the computation, so no page or file mixes units.

Quantity	Units offered	Conversion to internal unit
Matric potential h	cm, hPa, kPa	1 hPa = 1.019716 cm; 1 kPa = 10.19716 cm (ρ = 1000 kg/m³, g = 9.80665 m/s²)
Water content θ	cm³/cm³, %	1 % = 0.01 cm³/cm³
Conductivity K	cm/d, cm/h, mm/h, mm/d, m/d, m/s	to cm/d

Model parameters follow the units of the quantity they scale. θr and θs are reported like θ; α like 1/h (for example 1/kPa); h_m and h_clip like h. The shape parameters n, λ, σ and w_2 are dimensionless. pF_dry stays a decimal logarithm of the head in cm, as in the original PDI papers, and pore radii and diameters are always in µm.

The sign of the matric potential is ignored. Tensions may be entered as positive suction (100 cm) or as negative pressure head (−100 cm). Water contents in % are checked against the range 0 to 100, and fractions against 0 to 1, so a unit mix-up is caught before fitting.

3.3 Data input

Data can be typed into a table or imported from an Excel workbook. For an import the user gives the folder, file and sheet, and the first cell of three columns: sample names, matric potentials and water contents. Rows with the same sample name form one sample. Replicates need no averaging: every measured point enters the fit. Two tutorial data sets ship with the app in the Toturials folder and load with one click: Tutorial 1 fits each treatment with its replicates, Tutorial 2 fits each replicate separately. The same folder holds an Excel template for pressure plate data.

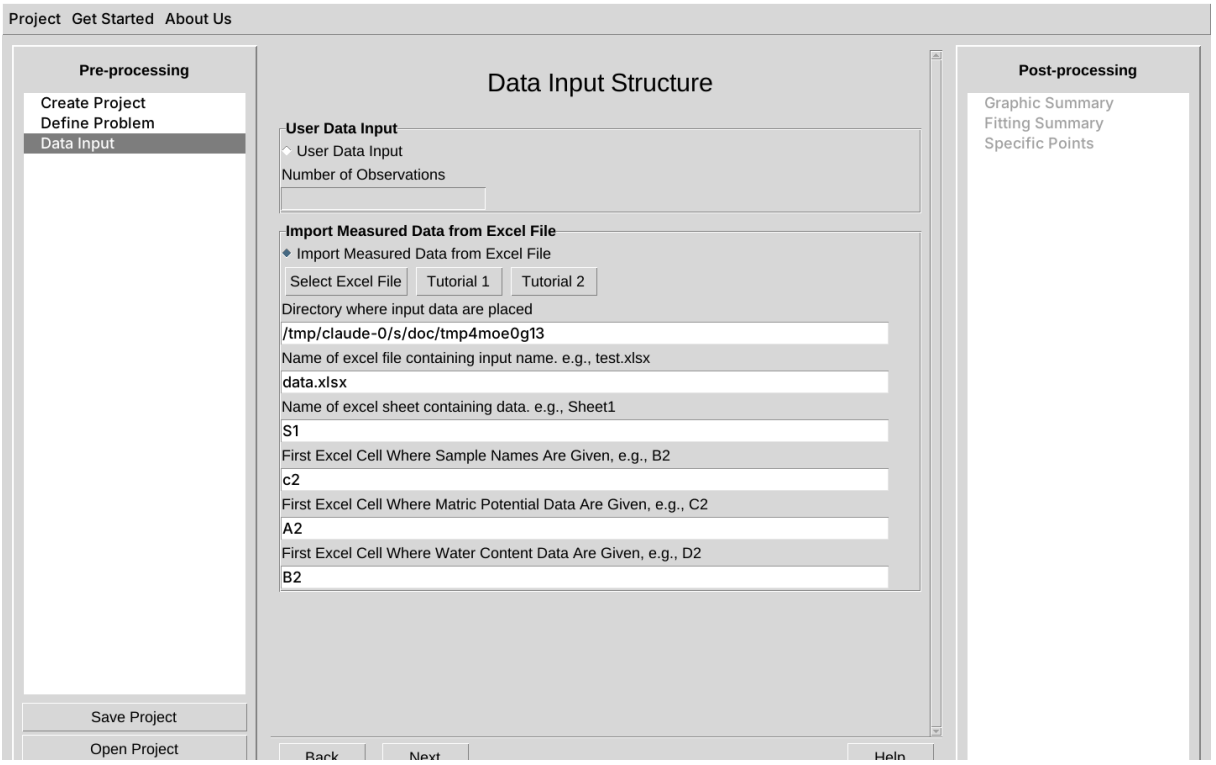


Figure 3.2. Data Input page. The workbook layout is described by the first cell of each data column.

3.4 Observations and weights

The Observations page lists the data as read. Values can be corrected, rows added or removed, and a weight assigned to each point. Decimal commas are accepted. Changing a unit on this page converts the listed values instead of reinterpreting them.

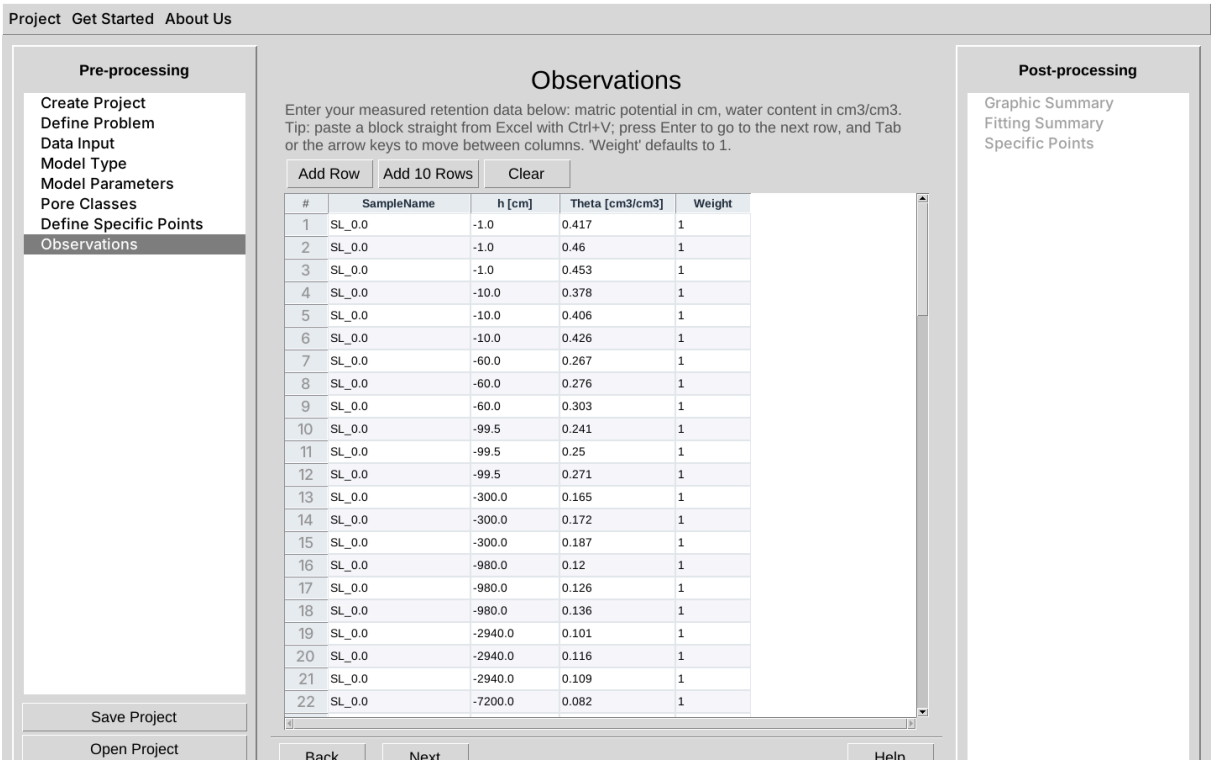


Figure 3.3. Observations page. Each row is one measurement. The weight column defaults to 1.

Weights are w = 1/σ, where σ is the standard deviation of the measurement error in θ. The fit minimizes Σ w_i^2 $(\theta_i - \theta(h_i))^2$. Note that the original RETC program uses w = $1/\sigma^2$ in the same position; values taken from RETC input files must be converted. A weight of 0 removes a point from the fit and from the degrees of freedom. Negative weights are rejected. Equal weights give an ordinary unweighted fit.

4. Retention models

The app offers six parametric models. Four are classical capillary models: van Genuchten, Brooks–Corey, Kosugi and the bimodal model of Durner. Two are the PDI variants of the unimodal and bimodal van Genuchten models, which add water adsorbed on particle surfaces and reach zero water content at oven dryness. All six share one structure: a saturation function between 0 and 1, scaled between a lower and an upper water content.

$$\theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h)$$

Here h is the matric potential expressed as a positive suction head, θs the saturated water content, θr the residual water content, and Se the effective saturation. Every model is also a statement about the pore system. Under the capillary bundle concept, a pore of equivalent radius r drains at the head given by the Young–Laplace equation (Section 8), so the derivative of Se with respect to log h is a pore-size distribution. This link is what allows the app to derive pore sizes (Section 8) and hydraulic conductivity (Section 9) from a fitted retention curve.

Model	Original reference	Parameters	Free parameters	Typical use
van Genuchten (VG)	van Genuchten (1980)	θr, θs, α, n	4	Default for most mineral soils
Brooks–Corey (BC)	Brooks and Corey (1964)	θr, θs, α, λ	4	Coarse, well-sorted soils with a sharp air entry
Kosugi (KOS)	Kosugi (1994, 1996)	θr, θs, h_m, σ	4	Physically interpretable lognormal pore sizes
Bimodal VG	Durner (1994)	θr, θs, α_1, n_1, w_2, α_2, n_2	7 (θr fixed by default)	Structured or aggregated soils with two pore systems
PDI van Genuchten	Peters (2013); Iden and Durner (2014); Peters et al. (2024)	θr, θs, α, n, pF_dry	5	Data reaching the dry range (pF > 4.2)
PDI bimodal	Peters (2013); Peters et al. (2024)	θr, θs, α_1, n_1, w_2, α_2, n_2, pF_dry	8 (θr, pF_dry fixed by default)	Structured soils with dry-range data

The default start values and bounds are wide on purpose, so that the same settings work for sands, loams, clays and peats. They can be changed on the Model Parameters page.

Parameter	Unit (internal)	Start	Lower bound	Upper bound
θr	cm^3/cm^3	0.02	0	0.25
θs	cm^3/cm^3	0.45	0.10	0.95
α, α_1	1/cm	0.1	10^{-4}	2
α_2	1/cm	0.03	10^{-4}	2
n, n_1, n_2	–	1.56, 1.7, 1.8	1.01	15
w_2	–	0.5	0	1
λ (BC)	–	0.9	0.1	15
h_m (KOS)	cm	100	0.1	10^6
σ (KOS)	–	1.5	0.1	5
pF_dry (PDI)	$\log_{10}$ cm	6.8	5.0	7.8

4.1 van Genuchten (1980)

Original reference. van Genuchten, M.Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. Soil Science Society of America Journal 44(5): 892–898.

The van Genuchten model is the most widely used retention function in soil physics. It describes a smooth sigmoid curve in log h, with a gradual air entry and an asymptotic approach to θr.

$$S_e(h) = \left[1 + (\alpha h)^n\right]^{-m}, \qquad m = 1 - \frac{1}{n}, \quad n > 1$$

- **α [1/h]** scales the head axis. Its inverse is roughly the head at which the soil starts to drain; a larger α means larger dominant pores. Coarse sands have α near 0.1 1/cm, clays near 0.005 1/cm.
- **n [–]** controls the width of the pore-size distribution. Large n (3 to 10) gives a steep curve and a narrow, well-sorted pore system; n close to 1 gives a flat curve and a wide distribution.
- **m = 1 − 1/n** is the Mualem restriction. van Genuchten introduced it because it gives a closed-form expression for the Mualem conductivity model (Section 9). The app always uses it, so K(h) is analytic and consistent with θ(h).

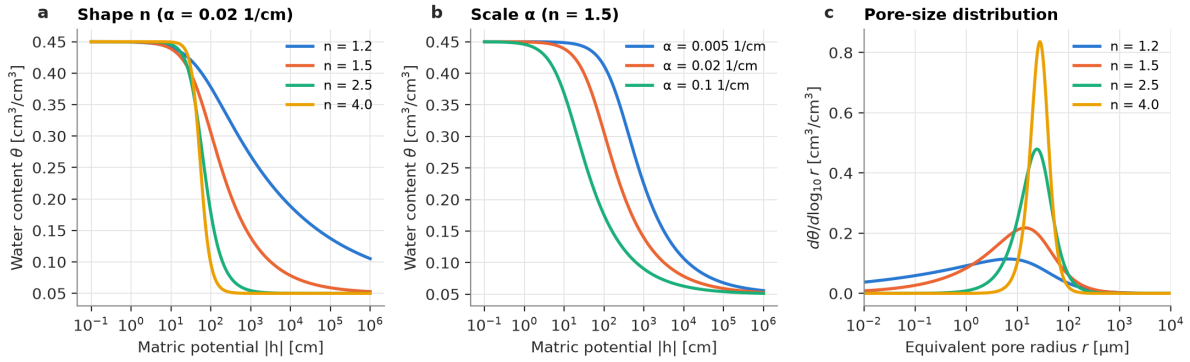


Figure 4.1. van Genuchten model. (a) n sets the steepness; (b) α shifts the curve along the head axis; (c) the volume-based pore-size distribution narrows as n grows. θr = 0.05, θs = 0.45.

The model has two known weaknesses. First, θr is a fitting parameter, not a measured property: it is the water content the curve approaches, and the model cannot reach zero water content at finite suction. Fits to data that end near pF 4.2 often return θr values that have no physical meaning in the dry range. Second, for n below about 2 the Mualem conductivity drops sharply near saturation (Ippisch et al., 2006). The app offers h_clip (Section 9) to handle this second point.

Use it as the default for most mineral soils measured in the pressure plate or HYPROP range (pF 0 to 4.2).

4.2 Brooks and Corey (1964)

Original reference. Brooks, R.H. and Corey, A.T. (1964). Hydraulic properties of porous media. Hydrology Paper No. 3, Colorado State University, Fort Collins, CO.

Brooks and Corey described drainage of porous media as a power law beyond a distinct air-entry (bubbling) pressure. The soil stays saturated until the largest pores begin to drain, then Se falls as a power of h.

$$S_e(h) = \begin{cases} 1, & h \le h_b = 1/\alpha \\ (\alpha h)^{-\lambda}, & h > h_b \end{cases}$$

- **α [1/h]** is the inverse of the air-entry head h_b. In the app it is labeled alpha, so it shares units with the van Genuchten α.
- **λ [–]** is the pore-size distribution index (named b in the code). Large λ means uniform pores; small λ means a wide range.

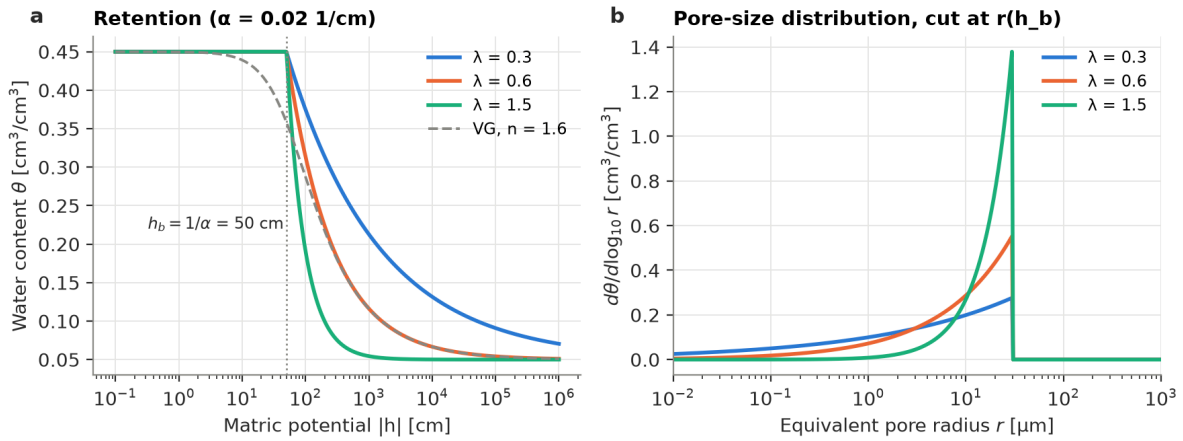


Figure 4.2. Brooks–Corey model with h_b = 50 cm. (a) The curve is flat up to h_b and follows a power law beyond; the dashed VG curve shows the smooth alternative. (b) The pore-size distribution ends abruptly at the radius that drains at h_b, because no pore is larger.

The kink at h_b makes the model a good description of well-sorted sands and glass beads, where a clear air entry is observed. It also has two practical costs. The derivative dS/dh is discontinuous at h_b, which slows the convergence of numerical flow models, and the pore-size distribution has a sharp edge that is rarely real in field soils. For the Mualem conductivity the model gives a simple power law, K = Ksat · Se^(τ + 2 + 2/λ) (Section 9).

Use it for coarse, well-sorted materials with a sharp air entry, or when results must be compared with studies that used Brooks–Corey parameters.

4.3 Kosugi (1996)

Original references. Kosugi, K. (1994). Three-parameter lognormal distribution model for soil water retention. Water Resources Research 30(4): 891–901. Kosugi, K. (1996). Lognormal distribution model for unsaturated soil hydraulic properties. Water Resources Research 32(9): 2697–2703.

Kosugi started from the pore system instead of the curve. He assumed that pore radii follow a lognormal distribution. Through Young–Laplace, the heads at which pores drain are lognormal too, and the retention function becomes a complementary error function.

$$S_e(h) = \frac{1}{2}\,\mathrm{erfc}\left[\frac{\ln(h/h_m)}{\sqrt{2}\,\sigma}\right]$$

- **h_m [h]** is the median head: the head at which half of the drainable pore volume has drained (Se = 0.5). It corresponds to the median pore radius r_m.
- **σ [–]** is the standard deviation of ln r (and of ln h). Small σ means uniform pores.

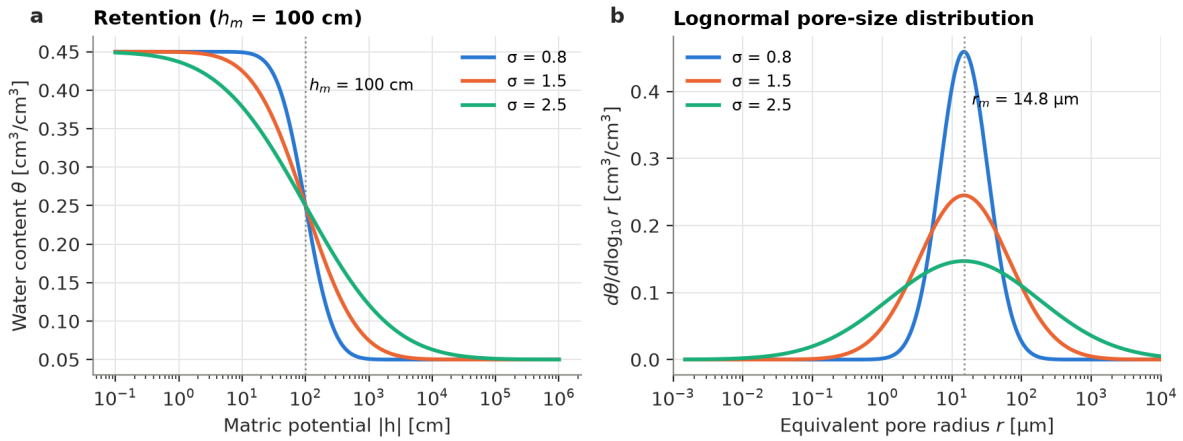


Figure 4.3. Kosugi model with h_m = 100 cm. (a) All curves pass through Se = 0.5 at h_m. (b) The pore-size distribution is a Gaussian in log r, centered on the median radius r_m = 14.8 µm.

Each parameter has a direct physical meaning, which makes the model useful for comparing treatments: a shift of h_m is a shift of the median pore size, a change of σ is a change of pore-size spread. The Mualem integral has a closed form, so K(h) is analytic. The app implements the three-parameter form of Kosugi (1996), with the head offset of the 1994 paper set to zero.

Use it when the pore-size distribution itself is of interest, for example to quantify how a treatment changes the median pore size.

4.4 Bimodal van Genuchten (Durner, 1994)

Original reference. Durner, W. (1994). Hydraulic conductivity estimation for soils with heterogeneous pore structure. Water Resources Research 30(2): 211–223.

Many soils have two pore systems: large inter-aggregate pores (cracks, biopores, packing voids) and small intra-aggregate or textural pores. A single sigmoid cannot follow the resulting double S-shape. Durner wrote the saturation as a weighted sum of van Genuchten functions, one per pore system. The app uses two.

$$S_e(h) = w_1\left[1 + (\alpha_1 h)^{n_1}\right]^{-m_1} + w_2\left[1 + (\alpha_2 h)^{n_2}\right]^{-m_2}, \qquad w_1 = 1 - w_2, \quad m_i = 1 - 1/n_i$$

- **α₁, n₁** describe the first pore system, **α₂, n₂** the second. Neither is forced to be the coarse one; the fit decides.
- **w₂ [–]** is the fraction of the drainable pore volume (θs − θr) that belongs to the second system.

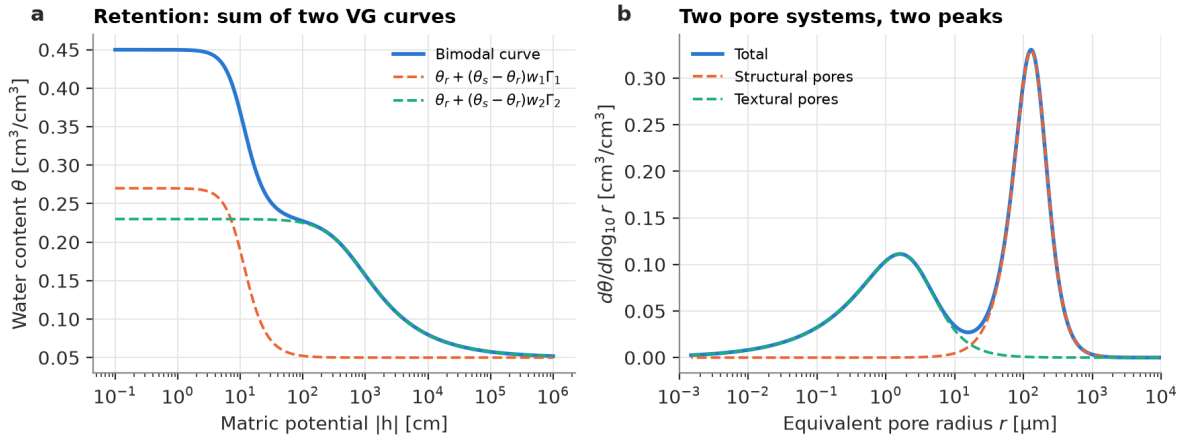


Figure 4.4. Bimodal model with α₁ = 0.1 1/cm, n₁ = 3, α₂ = 0.002 1/cm, n₂ = 1.6, w₂ = 0.45. (a) The curve is the sum of two VG components; (b) each component contributes one peak to the pore-size distribution.

The extra flexibility has a price. Seven parameters need data on both shoulders of the curve, and with typical pressure plate data (8 to 12 heads) the second mode is often not identifiable. The app therefore fixes θr by default. If the data are unimodal, the fit can drive w₂ to 0 or 1 or make the two modes coincide; the covariance matrix then becomes singular, and the app reports no standard errors and gives a warning instead of misleading intervals (Section 6). Compare models with the Akaike information criterion (Section 4.7) before choosing the bimodal form.

Use it for aggregated, structured or compacted soils whose data show two distinct drainage steps.

4.5 PDI van Genuchten (Peters, 2013; Iden and Durner, 2014)

Original references. Peters, A. (2013). Simple consistent models for water retention and hydraulic conductivity in the complete moisture range. Water Resources Research 49(10): 6765–6780. Iden, S.C. and Durner, W. (2014). Comment on “Simple consistent models for water retention and hydraulic conductivity in the complete moisture range” by A. Peters. Water Resources Research 50(9): 7530–7534. Peters, A., Durner, W. and Iden, S.C. (2024). The PDI model system for parameterizing soil hydraulic properties. Vadose Zone Journal 23(4): e20338.

Classical models treat θr as a constant floor. Real soils keep losing water as they dry, down to zero at oven dryness (about pF 6.8 to 7). The PDI model (named after Peters, Durner and Iden) splits the water into a capillary part, held in pores, and a non-capillary part, held as adsorbed films and in corners. Both reach zero at the oven-dry head h₀.

$$\theta(h) = (\theta_s - \theta_r)\, S_c(h) + \theta_r\, S_{nc}(h), \qquad h_0 = 10^{pF_{dry}}$$

$$S_c(h) = \frac{\Gamma(h) - \Gamma(h_0)}{1 - \Gamma(h_0)}, \qquad \Gamma(h) = \left[1 + (\alpha h)^n\right]^{-m}$$

$$S_{nc}(h) = \frac{\ln(h_0/h) - b \ln\left[1 + (h_a/h)^{1/b}\right]}{\ln(h_0/h_a)}, \qquad b = 0.2 \ln 10$$

- **θr** changes meaning: it is the maximum adsorbed water content, the water held by the non-capillary term at saturation. It is no longer a residual value.

- **α, n** describe the capillary pore system, as in the van Genuchten model.
- **pF_dry** is $\log_{10}$ of h_0 in cm. The default is 6.8 (oven dryness at about 105 °C). Beyond h_0 the app sets θ = 0.
- **h_a** is not a free parameter. It is the head where the capillary saturation Sc equals 0.75, so the adsorbed water starts to drain once most capillary pores are empty. b is a fixed smoothing constant.

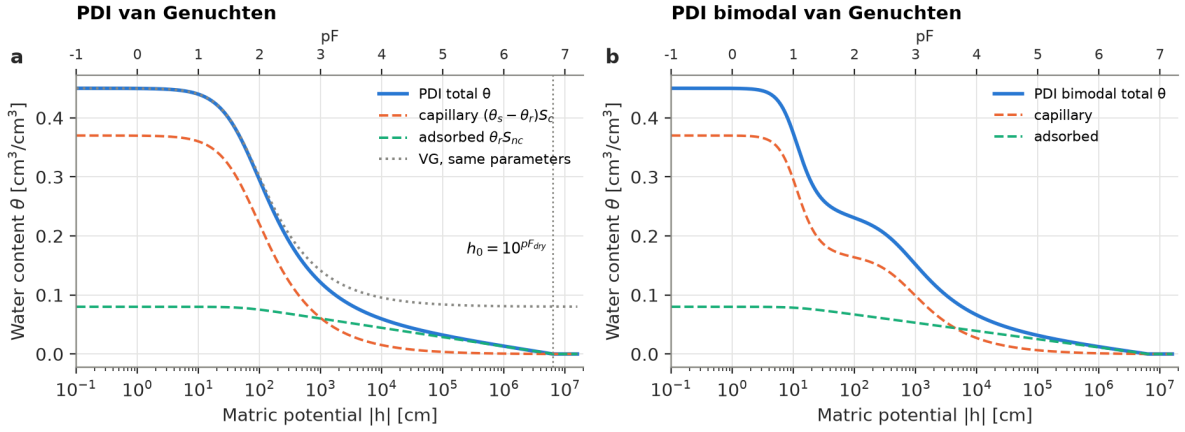


Figure 4.5. (a) PDI van Genuchten model with θr = 0.08, θs = 0.45, α = 0.02 1/cm, n = 1.6, pF_dry = 6.8. The capillary part (orange) drains like a VG curve; the adsorbed part (green) decreases linearly in log h to zero at h_0. The classical VG curve (dotted) stays at θr. (b) The PDI bimodal model adds a second capillary mode.

The capillary part keeps the shape of the van Genuchten curve, so the two models agree in the wet range. They differ beyond pF 3, where PDI follows the near-linear decline of θ against log h that is observed in dry soils. The PDI model also supplies film and vapor conductivity (Section 9), which dominate flow in dry soil.

With data that end at pF 4.2, pF_dry cannot be estimated; its fit runs to a bound and the app flags it AT BOUND. Fix it at 6.8 in that case, as the app does for the bimodal PDI model.

Use it when data reach the dry range (dew-point or vapor sorption measurements), when evaporation or drying processes are modeled, or when a physically consistent θ = 0 at oven dryness is needed.

4.6 PDI bimodal van Genuchten

Original references. Peters (2013); Peters, Durner and Iden (2024), as above. The capillary function is the bimodal function of Durner (1994).

The PDI bimodal model replaces Γ(h) in the PDI equations by the bimodal saturation of Section 4.4. Here h_a is found numerically as the root of Sc(h_a) = 0.75, by bisection in log h. All other terms are as in Section 4.5. With eight parameters the model needs rich data; by default θr and pF_dry are fixed, leaving six to fit.

Use it for structured soils with data that reach the dry range.

4.7 Choosing a model

Goodness of fit alone rarely decides between models, because most of them fit typical data within measurement error. Figure 4.6 shows all six fitted to the same sample (SL_0.0 of the

tutorial data, 27 points from 1 to 12350 cm).

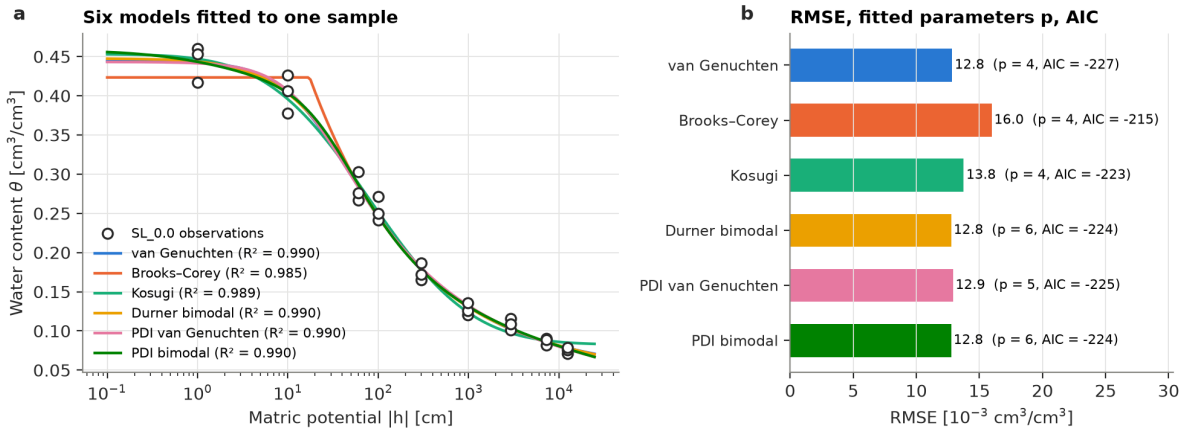


Figure 4.6. Six models fitted to one sample. (a) Fitted curves; (b) root mean square error, number of fitted parameters p and Akaike information criterion (lower is better).

Four models reach $R^2 = 0.990$, and the extra parameters of the bimodal and PDI models buy almost no reduction in RMSE. The Akaike information criterion ($AIC = n \ln(RSS/n) + 2p$) penalizes each added parameter and ranks van Genuchten best for this sample. Brooks–Corey fits worst: the data show a smooth air entry, not a kink. The app reports AIC and BIC for every fit, so the same comparison can be made for any data set.

Practical guidance:

1. Start with van Genuchten.
2. Use Kosugi when pore-size statistics (median and spread) are the target.
3. Use Brooks–Corey only when the data show a sharp air entry.
4. Use a bimodal model only when the residuals of the unimodal fit show a systematic double S-shape and the AIC improves.
5. Use a PDI model when data extend beyond pF 4.2 or when K in dry soil matters.

5. Parameter estimation

Parameters are estimated by weighted nonlinear least squares, separately for each sample, using every observation. The approach is the same as in RETC (van Genuchten et al., 1991), with a more robust optimizer and full uncertainty output.

5.1 Model choice and parameter setup

The Model Type page lists the six models, shows the equation, the parameter meaning and the original reference of the selected model, and draws an example curve.

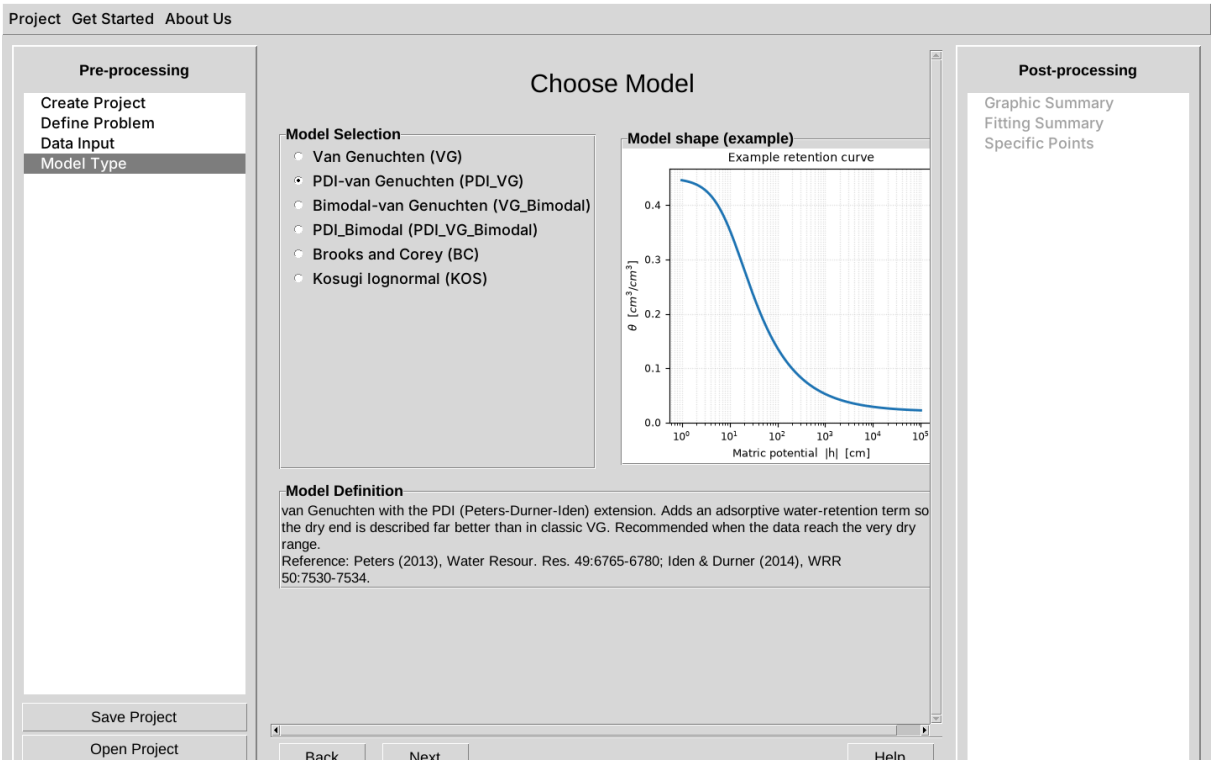


Figure 5.1. Model Type page. The PDI van Genuchten model is selected.

The Model Parameters page holds one row per parameter: unit, a Fitted? check box, start value, and lower and upper bound, all in the user's units. Unticking Fitted? fixes a parameter at its start value. Each model keeps its own table, so switching models and back does not lose entries. When K(h) is requested, a second table holds the conductivity inputs (Section 9).

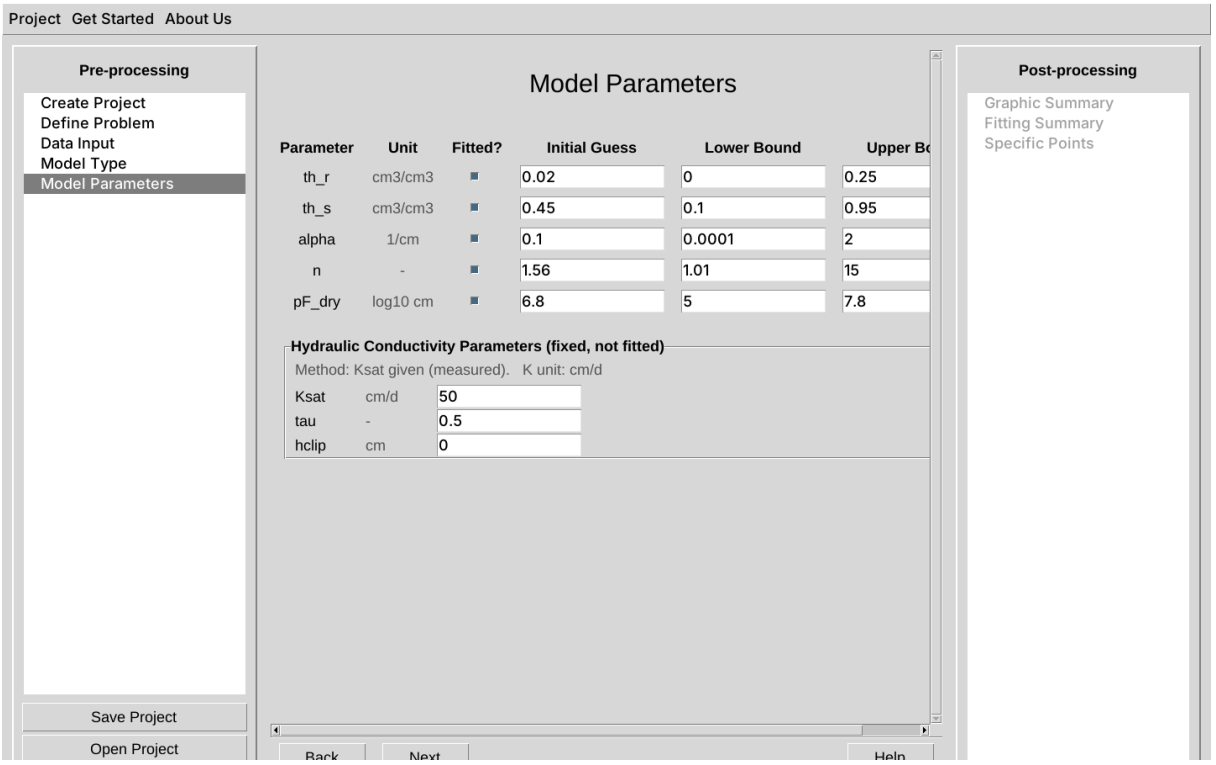


Figure 5.2. Model Parameters page for PDI van Genuchten with K prediction from a measured Ksat.

The page checks the table before moving on: start values must be numbers, each lower bound must be below its upper bound, n must exceed 1, and at least one parameter must be free. With fewer data points than free parameters the fit is refused with a clear message.

5.2 Objective function

For a sample with N observations (h_i, θ_i) and parameter vector p, the app minimizes

$$\Phi(\mathbf{p}) = \sum_{i=1}^{N} w_i^2 \left[\theta_i - \theta(h_i; \mathbf{p})\right]^2$$

subject to the bounds. With $w_i = 1/\sigma_i$ this is the maximum likelihood estimate for independent Gaussian errors. Without weights all $w_i = 1$. Replicates are entered as separate observations, so a head measured three times counts three times, and samples with unequal replicate numbers are handled correctly.

5.3 Optimizer

The minimization runs in two stages, both from the lmfit library (Newville et al., 2014):

1. **Nelder–Mead simplex** (Nelder and Mead, 1965) from the start values, with moderate tolerances (x 10^{-3}, objective 10^{-6}) and at most 300 evaluations per free parameter. The simplex needs no derivatives and moves reliably from a rough start into the basin of the optimum.
2. **Trust-region reflective least squares** (scipy least_squares) from the simplex result. It converges quadratically near the optimum, respects the bounds, and returns the Jacobian needed for the covariance matrix.

The two-stage scheme removes most of the sensitivity to start values that single-stage Levenberg–Marquardt fits show, especially for the bimodal models.

5.4 Goodness of fit

For each sample the app reports:

Statistic	Definition	Note
R^2	$1 - \Sigma(\theta_i - \hat{\theta}_i)^2 / \Sigma(\theta_i - \bar{\theta})^2$	Unweighted, 3 decimals
RMSE	$\sqrt{[\Sigma(\theta_i - \hat{\theta}_i)^2 / N]}$	In the user's θ unit
Reduced chi-square	$\Phi / (N - p)$	Estimate of the error variance; in θ unit² when unweighted
AIC	$N \ln(\Phi/N) + 2p$	For model comparison on the same data
BIC	$N \ln(\Phi/N) + p \ln N$	Penalizes parameters more for large N
N, p	Observations used, free parameters	Points with weight 0 are excluded

A parameter that ends within 0.1% of a bound is marked AT BOUND. Its standard error and confidence interval are then unreliable, because the minimum lies on the boundary, not in the

interior where the linear theory of Section 6 applies. A bound hit usually means the data do not constrain that parameter (as pF_dry with pressure plate data) or that the bound is too narrow.

6. Uncertainty: 95% confidence intervals

Every estimate the app reports carries a two-sided 95% confidence interval. The intervals come from the linearized (asymptotic) theory of nonlinear regression (Bates and Watts, 1988; Seber and Wild, 1989), applied in one consistent way to parameters and to every quantity derived from them. A Monte Carlo test (Section 6.4) shows that the stated 95% is met.

6.1 Parameter covariance

At the optimum p̂ the trust-region step returns the Jacobian J of the weighted residuals. The covariance matrix of the parameters is

$$\mathbf{C} = s^2 \left(\mathbf{J}^{\mathsf{T}} \mathbf{J} \right)^{-1}, \qquad s^2 = \frac{\Phi\left(\widehat{\mathbf{p}}\right)}{N - p}$$

where s² is the reduced chi-square. Scaling by s² makes the intervals reflect the actual scatter of the data, as in RETC. The standard error of parameter j is √C_jj, and the interval is

$$\widehat{p}_j \pm t_{0.975,\, N-p} \sqrt{C_{jj}}$$

with the Student t quantile for N − p degrees of freedom. For small data sets t is noticeably larger than 1.96 (2.07 for 27 points and 4 parameters, 2.57 for 9 points and 4 parameters), which keeps the intervals honest.

The Fitting Summary also lists the correlation coefficients C_jk / √(C_jj C_kk) above 0.1 in magnitude. Strong correlations, for example between α and n, explain why single parameters can be uncertain while the curve itself is well defined.

6.2 Derived quantities: the delta method

For any quantity g(p) computed from the parameters (θ at a head, plant-available water, a pore-class volume, log K), the app propagates C by the delta method:

$$\mathrm{Var}\,[\,g\,] \approx \nabla g^{\mathsf{T}}\, \mathbf{C}\, \nabla g, \qquad g\left(\widehat{\mathbf{p}}\right) \pm t_{0.975,\, N-p} \sqrt{\mathrm{Var}\,[\,g\,]}$$

The gradient ∇g is computed by central finite differences with a relative step of 10^{-6}. One routine serves all quantities, so every interval in the app is computed the same way. The full covariance is used, including off-diagonal terms, so correlated parameters do not inflate or deflate the result.

Two transformations keep intervals realistic for quantities that span orders of magnitude. For h at a given water content, and for K(h), the interval is computed for $\log_{10} h$ and $\log_{10} K$ and transformed back, so it is asymmetric on a linear scale and never negative.

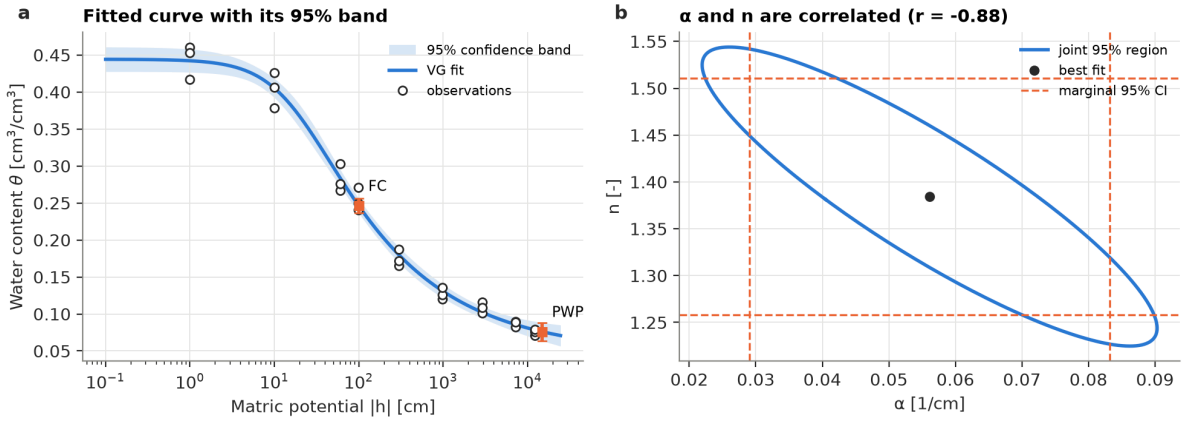


Figure 6.1. Sample SL_0.0, van Genuchten fit. (a) The 95% confidence band of the fitted curve and the intervals of θ at FC (100 cm) and PWP (15000 cm). The band is narrow where data are dense and widens at the dry end. (b) The joint 95% confidence region of α and n (ellipse) and their marginal intervals (dashed). The correlation of −0.88 tilts the ellipse.

The band in Figure 6.1a is the confidence interval of the fitted mean curve, not a prediction interval for new measurements. It answers “how well is the curve known”, which is the right question for comparing treatments. New single measurements scatter around the curve with the additional variance s^2.

6.3 When intervals are not reported

If the data cannot determine all free parameters, J^TJ is singular or nearly so and C is meaningless. The app tests this on the correlation matrix, which is independent of units: if its smallest eigenvalue is below 10^{-10}, or any variance is zero or not finite, no standard errors or intervals are reported, and the goodness-of-fit block carries a warning to fix parameters or choose a simpler model. Typical causes are a bimodal model fitted to unimodal data, or pF_dry fitted without dry-range data. Parameters that end on a bound are flagged AT BOUND (Section 5.4).

6.4 Verification by Monte Carlo

The coverage of the intervals was tested by simulation. Synthetic data were generated from a known van Genuchten curve (θr = 0.046, θs = 0.444, α = 0.056 1/cm, n = 1.38) at the 9 heads of the tutorial sample, three replicates each, with Gaussian noise of standard deviation 0.0128 cm³/cm³ (the RMSE of the real fit). Each of 400 data sets was fitted, and the share of intervals containing the true value was counted.

Quantity	Coverage of the nominal 95% interval
θr	93.8%
θs	93.5%
α	93.2%
n	94.8%
θ at FC (100 cm)	95.2%
θ at PWP (15000 cm)	94.5%
PAWC	95.5%
h at θ = 0.20	96.5%

All values lie within 93% to 97%, close to the nominal 95% (the Monte Carlo error with 400 runs is about ±1.1 percentage points). Slight under-coverage of single parameters is expected from the curvature of the model; derived quantities are close to or above nominal. The test script is part of the verification set (Section 11).

7. Specific points: field capacity, wilting point and plant-available water

The fitted curve is read at user-defined heads and water contents, and each value is reported with its 95% interval. The defaults are field capacity (FC) at 100 cm (pF 2.0) and permanent wilting point (PWP) at 15000 cm (pF 4.2). Both can be changed, and further points can be added.

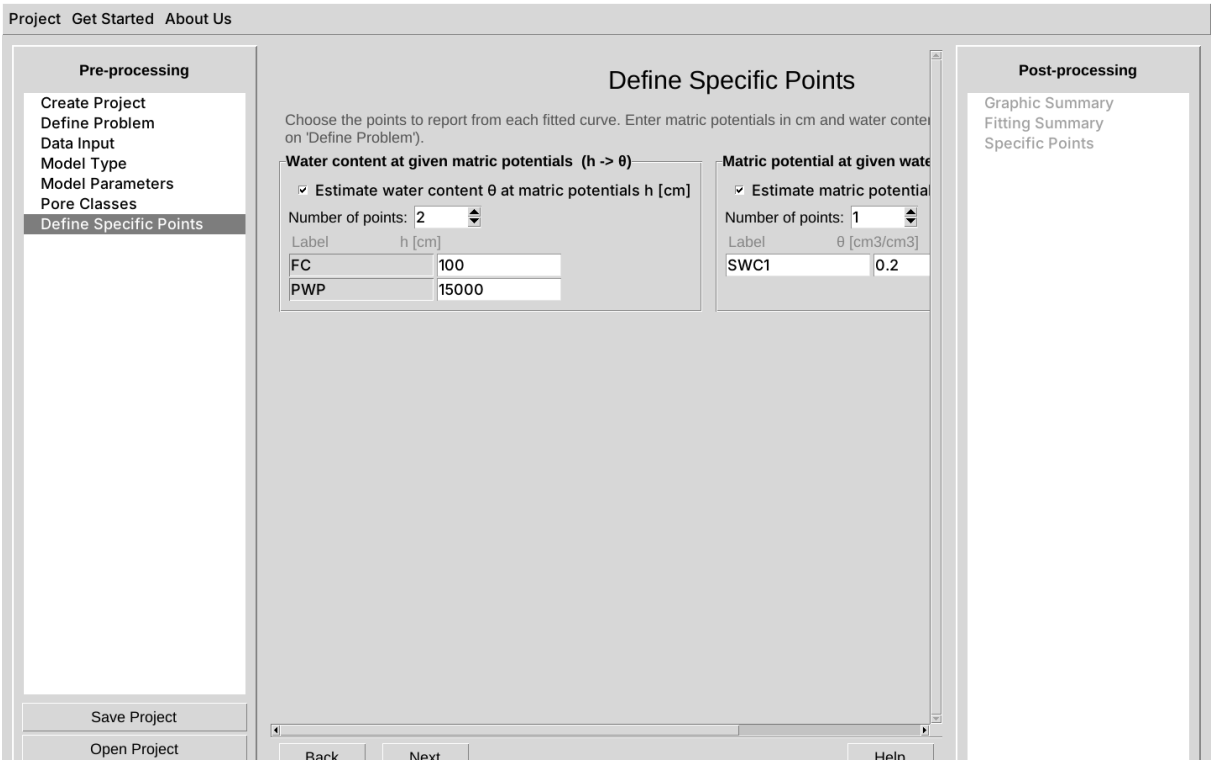


Figure 7.1. Define Specific Points page. Left: water content at given matric potentials. Right: matric potential at given water contents.

Three kinds of point are computed:

Point	Definition	95% interval
θ at h	θ(h) of the fitted curve	Delta method, symmetric
PAWC	θ(h_FC) − θ(h_PWP), reported when points labeled FC and PWP exist	Delta method on the difference, full covariance
h at θ	The head where θ(h) = θ*, found by bisection in $\log_{10}$ h between 10^{-8} and 10^{7} cm	Delta method on $\log_{10}$ h, asymmetric in h

The plant-available water content needs care. θ(FC) and θ(PWP) come from the same parameters and are correlated. Treating them as independent and adding their variances would misstate the interval. The app computes the variance of the difference directly, so the correlation is included.

The FC head is a convention, not a soil constant. Common choices are 60 cm (pF 1.8, German KA5), 100 cm (pF 2.0) and 330 cm (pF 2.5, USDA). Set the value that matches the reporting standard of the study.

For sample SL_0.0 with the van Genuchten fit:

Point	Value	95% interval	Unit
θ at FC (100 cm)	0.246	0.237 to 0.256	cm³/cm³
θ at PWP (15000 cm)	0.076	0.064 to 0.088	cm³/cm³
PAWC	0.170	0.153 to 0.187	cm³/cm³
h at θ = 0.20	206	173 to 246	cm

The results appear in the Specific Points page, in Specific Points.txt, in the Excel files and in two bar charts that compare all samples.

8. Pore-size distribution and pore classes

A retention curve is an indirect measurement of the pore system. Read through the capillary law, each step of drainage empties pores of one size, so the slope of the curve is a pore-size distribution. The app derives this distribution from the fitted model and sums it into pore classes with confidence intervals.

8.1 From matric potential to pore size

The Young–Laplace equation gives the radius r of a cylindrical pore that drains at matric head h, for a fully wetting liquid (contact angle 0):

$$r = \frac{2\,\sigma_w(T)}{\rho_w(T)\,g\,h}, \qquad d = 2r = \frac{4\,\sigma_w(T)}{\rho_w(T)\,g\,h}$$

Surface tension σ_w and density ρ_w depend on temperature T; the app uses 20 °C, which gives r [µm] ≈ 1486 / h [cm] and d [µm] ≈ 2970 / h [cm]. Since r ∝ 1/h, each decade of head is one decade of pore size.

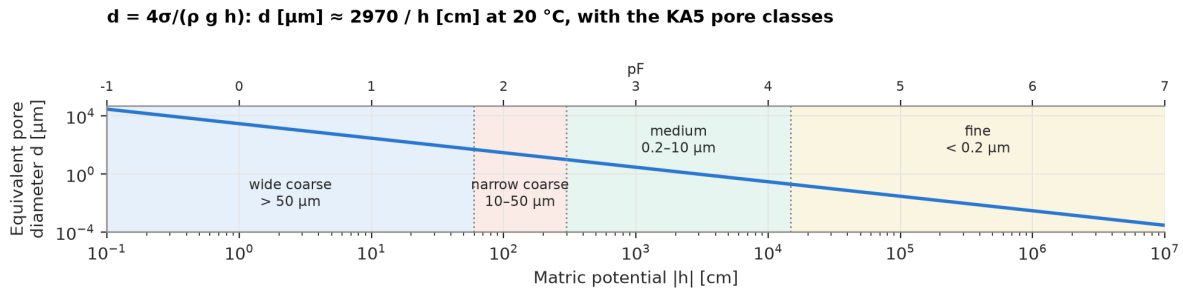


Figure 8.1. Equivalent pore diameter as a function of matric potential, with the four KA5 classes. Class limits at 50, 10 and 0.2 µm correspond to 59.4, 297 and 14840 cm (pF 1.77, 2.47 and 4.17).

The radius is an equivalent radius: the radius of a cylinder that would drain at the same head. Real pores are irregular, connected through narrower necks, and partly drain only when their neck empties. The derived distribution therefore describes the pore necks that control drainage, which is also what controls water flow.

8.2 The distribution function

The app reports the distribution per decade of radius. Because r ∝ 1/h, the derivative of the capillary saturation with respect to $\log_{10}$ r follows from the head derivative:

$$\frac{dS}{d\log_{10} r} = \ln(10)\, h \left|\frac{dS}{dh}\right|$$

Its integral over all radii is 1 for every model. Two forms are reported:

- **Normalized** dS/dlog₁₀ r [–]: the shape of the distribution. Two samples with the same pore-size spread but different porosity look alike.
- **Volume-based** dθ/dlog₁₀ r = (θs − θr) dS/dlog₁₀ r [cm³/cm³]: the volume of water held per decade of radius. Its area equals the drainable pore volume θs − θr. This is the form shown in the figures, because differences in total porosity between treatments stay visible.

The distribution is analytic for every model, so no numerical differentiation of data is involved. The models differ in what they imply:

Model	Shape of the implied distribution
van Genuchten	Skewed, single peak; long tail toward small pores when n is small
Brooks–Corey	Power law, cut off at the air-entry radius (no larger pores)
Kosugi	Lognormal: a Gaussian in log r, centered on r_m
Bimodal	Two peaks, one per pore system
PDI models	Capillary part only (Sc); adsorbed water is not pore water and is excluded

The pore-size distribution is a property of the fitted model. Beyond the range of the data it is an extrapolation of the model shape. Pores larger than about 1500 µm (h < 1 cm) or smaller than

about 0.1 µm (h > 15000 cm) are rarely measured in pressure plates, so the tails of the distribution should be read with that in mind.

8.3 Pore classes

Pore-size classes summarize the distribution into a few volumes that can be compared between soils and treatments. The default classes follow the German soil mapping guide KA5 (Ad-hoc-AG Boden, 2005), by equivalent diameter:

Class	Equivalent diameter	Head range (20 °C)	Function
Wide coarse pores	> 50 µm	0 to 59.4 cm	Rapid drainage, aeration
Narrow coarse pores	10 to 50 µm	59.4 to 297 cm	Slow drainage
Medium pores	0.2 to 10 µm	297 to 14840 cm	Plant-available water
Fine pores	< 0.2 µm	> 14840 cm	Water not available to plants

A pore with diameter between d_low and d_high is full at the head that drains d_high and empty at the head that drains d_low. The class volume is therefore read directly from the fitted curve:

$$V_{class} = \theta(h\left(d_{high}\right)) - \theta(h\left(d_{low}\right))$$

The finest class (d_low = 0) takes all water still held at its wet head, including residual or adsorbed water, as in the KA5 definition. With classes that cover 0 to ∞, the volumes add up exactly to θs. Each volume carries a delta-method 95% interval (Section 6).

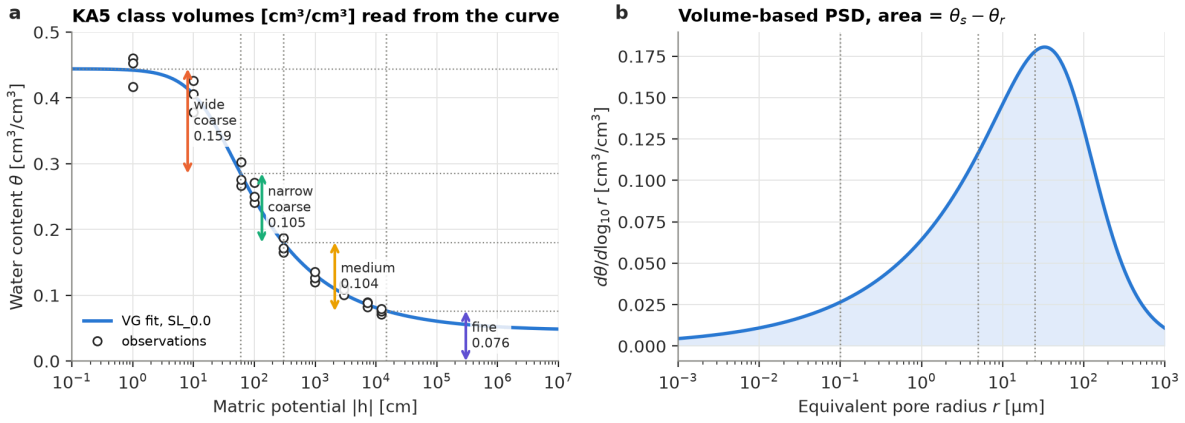


Figure 8.2. Sample SL_0.0, van Genuchten fit. (a) Class volumes are differences of θ between the class limits; they sum to θs = 0.444. (b) The volume-based pore-size distribution; dotted lines mark the class limits in radius (25, 5 and 0.1 µm).

The classes can be replaced on the Pore Classes page. Custom classes need a name and a lower and upper diameter in µm; inf marks the open upper end. The page checks that every class has d_low < d_high and shows the matching head range.

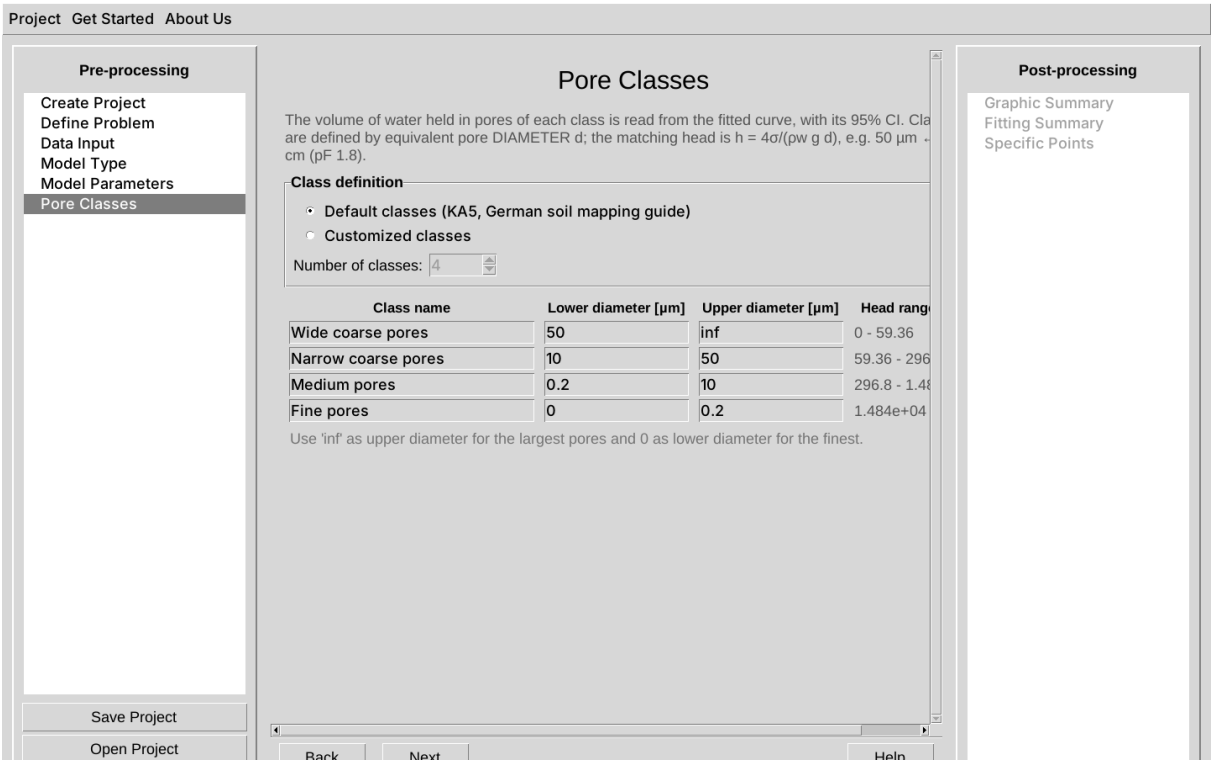


Figure 8.3. Pore Classes page with the KA5 defaults.

The pore-size distribution and the class volumes are shown in the Graphic Summary for all samples together.

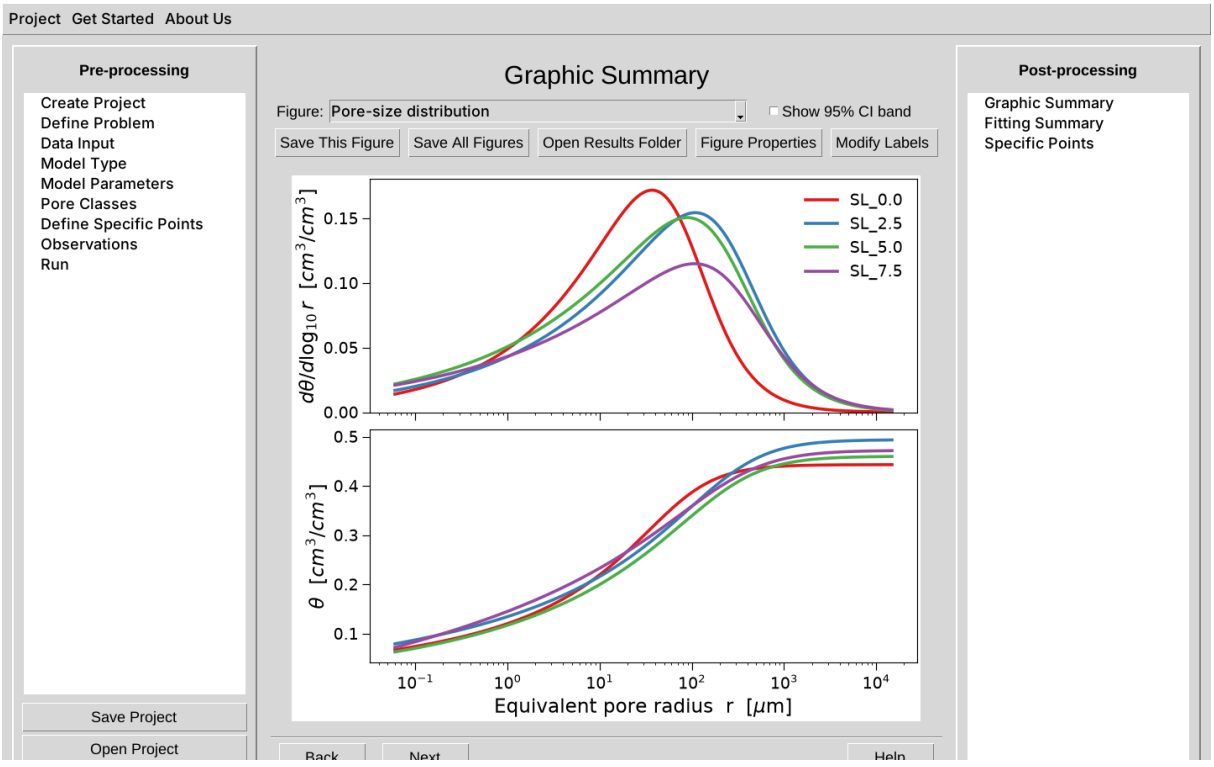


Figure 8.4. Graphic Summary, pore-size distribution view: volume-based distribution (top) and the retention curves on the radius axis (bottom) for the four tutorial samples.

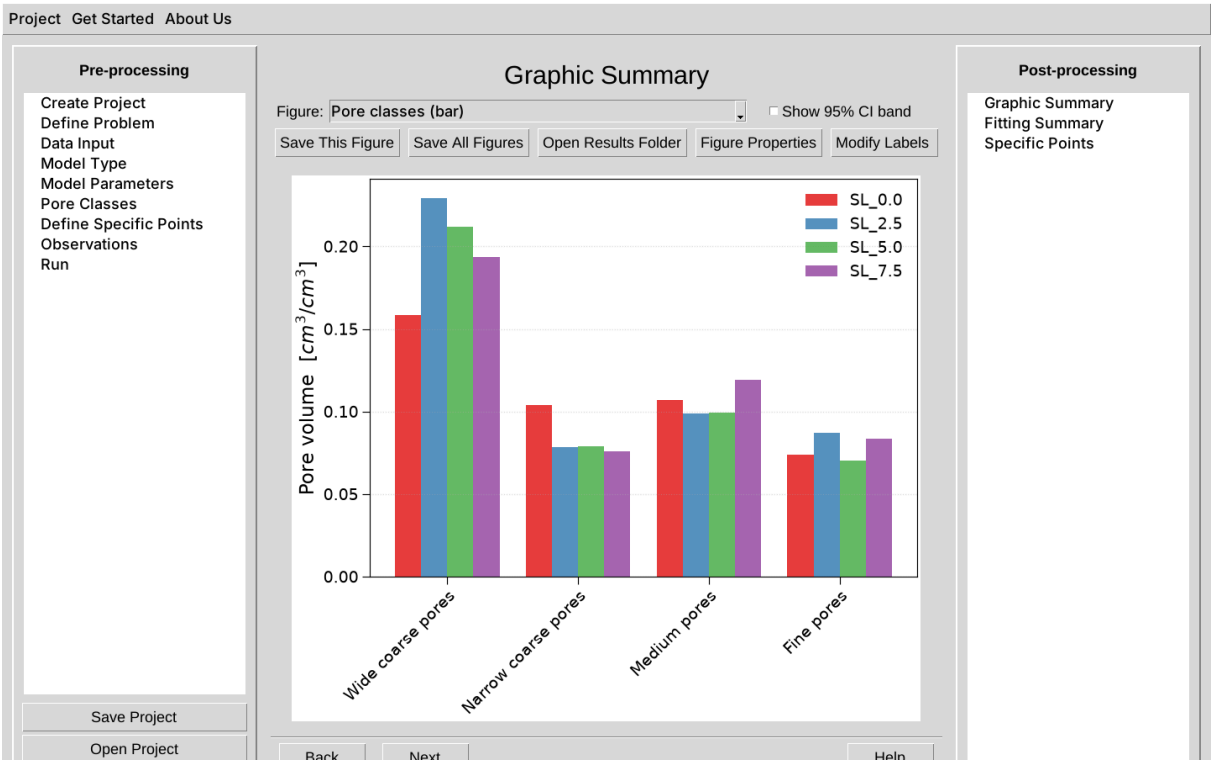


Figure 8.5. Graphic Summary, pore classes view. Error bars show the 95% intervals when the CI option is ticked.

9. Hydraulic conductivity predicted from the retention curve

Unsaturated conductivity is slow and costly to measure; retention data are not. Because both depend on the same pore system, K(h) can be predicted from the fitted retention curve. The app implements the capillary bundle model of Mualem (1976) for all six models, with either a measured saturated conductivity or one predicted from the retention curve itself (Peters et al., 2023). For the PDI models it adds film flow (Peters et al., 2021), and for every model optional isothermal vapor flow (Saito et al., 2006).

The option is ticked on the Define Problem page, where the method and the K unit are chosen. The values (Ksat or τ_s, τ, h_clip) are entered on the Model Parameters page and are held fixed during fitting: the retention fit is never influenced by the K settings.

9.1 Mualem's capillary model

Original reference. Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. Water Resources Research 12(3): 513–522.

Mualem pictured the soil as a bundle of capillaries cut into slices and randomly reconnected. Flow through each connection follows Poiseuille's law, which leads to

$$K(S_e) = K_s S_e^{\tau}\left[\frac{I(S_e)}{I(1)}\right]^2, \qquad I(S_e) = \int_0^{S_e} \frac{dS}{h(S)}$$

The integral weights each pore class by 1/h, that is by its radius. The tortuosity–connectivity exponent τ is 0.5 in Mualem's original analysis. Schaap and Leij (2000) found that τ fitted to many data sets is often negative (around −1); the app lets the user set it.

The integral has a closed form for every model in the app, so K is analytic and consistent with θ(h):

Model	Mualem integral I(Se)
van Genuchten (m = 1 − 1/n)	α [1 − (1 − $Se^{(1/m))}$m]
Brooks–Corey	αλ / (λ + 1) · Se^(1 + 1/λ), so K = Ks Se^(τ + 2 + 2/λ)
Kosugi	exp(σ²/2) / h_m · ½ erfc[$erfc^{-1}$(2Se) + σ/√2]
Bimodal	Weighted sum of the two VG integrals (Durner, 1994)
PDI models	Same integrals over the capillary saturation Sc, from the dry end h_0

The van Genuchten form is evaluated as −α expm1[m log1p(−Se^(1/m))]. This is algebraically identical but keeps full precision in dry soil, where the textbook form cancels to zero and K would be lost below about 10^{-16} of Ks.

9.2 Saturated conductivity: given or predicted

Ksat given. The user enters a measured Ks in any of the six K units. This is the recommended option whenever a measurement exists.

Ksat predicted. Peters et al. (2023) showed that the absolute conductivity, not only its relative shape, follows from the retention curve when the capillary bundle is scaled by water properties and one tortuosity factor:

$$K(S_c) = \beta \tau_s S_c^{\tau} \left[(\theta_s - \theta_r) \int_0^{S_c} \frac{dS}{h(S)} \right]^2, \qquad \beta = \frac{\sigma_w^2}{2 \mu_w \rho_w g}$$

Original reference. Peters, A., Hohenbrink, T.L., Iden, S.C., van Genuchten, M.Th. and Durner, W. (2023). Prediction of the absolute hydraulic conductivity function from soil water retention data. Hydrology and Earth System Sciences 27(7): 1565–1582.

Here μ_w is the dynamic viscosity of water. At 20 °C the app computes β = 2.67 × 10^{-4} m^3/s (2.31 × 10^7 cm^3/d) from its temperature functions for σ_w, μ_w and ρ_w. Peters et al. (2023) quote β = 3.04 × 10^{-4} m^3/s at 20 °C, so K predicted by the app is lower by a factor of 1.14 than with the published constant. This is small compared with the spread of τ_s between soils, but should be kept in mind when comparing with values from the paper. The factor τ_s lumps tortuosity and connectivity of the saturated pore network; Peters et al. (2023) found values around 0.1 across many soils, which is the default. The prediction is usually within about one order of magnitude of measured values. It is useful when no Ks is available, but it cannot see macropores, cracks or biopores that the retention data do not resolve.

9.3 Clipping the largest pores: h_clip

Models with a smooth air entry imply pores of unlimited size near h = 0. With n below about 2, the van Genuchten–Mualem K then falls by orders of magnitude between saturation and a few cm of suction (Ippisch et al., 2006). Iden et al. (2015) proposed to limit the largest pore by treating all heads below h_clip as h_clip in the Mualem integral. The app applies this to all

models when h_clip > 0 (default 0, off). A typical value is 1 cm, which caps the pore radius at about 1.5 mm.

9.4 Film and vapor flow

In dry soil, capillary flow stops and water moves as thin films on grain surfaces and as vapor.

Film conductivity (PDI models only). Peters et al. (2021) derived K_film from the physics of adsorbed films. It depends on the non-capillary saturation S_nc:

$$K_{film}(h) = c(T)\ \theta(h_{ref})\ h_a^{-a}\left(\frac{h_0}{h_a}\right)^{-a(1-S_{nc})}, \qquad a = 1.5,\ h_{ref} = 10^5\ \text{cm}$$

The factor c combines viscosity, the Hamaker constant, permittivity and temperature (c = 86.2 at 20 °C in the app's units). The film term needs pF_dry > 5, because its reference head lies at pF 5.

Original reference. Peters, A., Hohenbrink, T.L., Iden, S.C. and Durner, W. (2021). A simple model to predict hydraulic conductivity in medium to dry soil from the water retention curve. Water Resources Research 57(5): e2020WR029211.

Isothermal vapor conductivity (optional, all models). Following Saito et al. (2006), vapor moves by diffusion along the gradient of vapor density, which the matric potential controls through the Kelvin equation:

$$K_v(h) = \frac{\rho_{sv}(T)}{\rho_w} D_v(\theta)\ \frac{Mg}{RT} H_r(h), \qquad D_v = D_a \frac{(\theta_s - \theta)^{10/3}}{\theta_s^2}, \qquad H_r = \exp\left(-\frac{Mgh}{RT}\right)$$

The vapor diffusivity in soil uses the Millington and Quirk (1961) tortuosity model; ρ_sv is the saturated vapor density and H_r the relative humidity.

Original references. Saito, H., Šimůnek, J. and Mohanty, B.P. (2006). Numerical analysis of coupled water, vapor, and heat transport in the vadose zone. Vadose Zone Journal 5(2): 784–800. Millington, R.J. and Quirk, J.P. (1961). Permeability of porous solids. Transactions of the Faraday Society 57: 1200–1207.

The total conductivity is the sum K = K_cap + K_film + K_v.

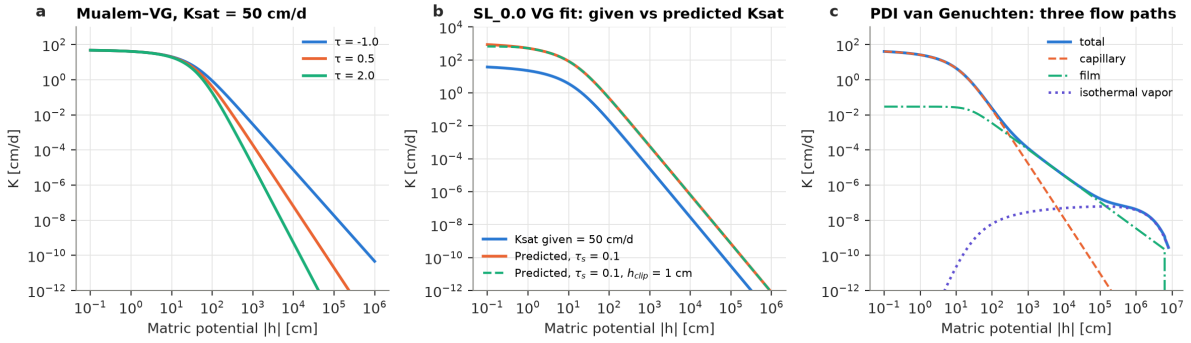


Figure 9.1. (a) Effect of τ on the Mualem–VG conductivity. (b) Sample SL_0.0: K with a measured Ks of 50 cm/d versus the Peters et al. (2023) prediction with τ_s = 0.1; clipping at 1 cm changes the wet end only. (c) PDI van Genuchten: capillary flow dominates the wet range, film flow takes over beyond about pF 3.5, and vapor flow near oven dryness.

9.5 Uncertainty of K

The 95% band of K(h) is computed by the delta method on $\log_{10} K$ (Section 6.2), from the covariance of the fitted retention parameters. Ks, τ_s, τ and h_clip are fixed inputs, so their own uncertainty is not included. The band thus shows how well the retention data constrain the shape of K(h), not the full uncertainty of the prediction.

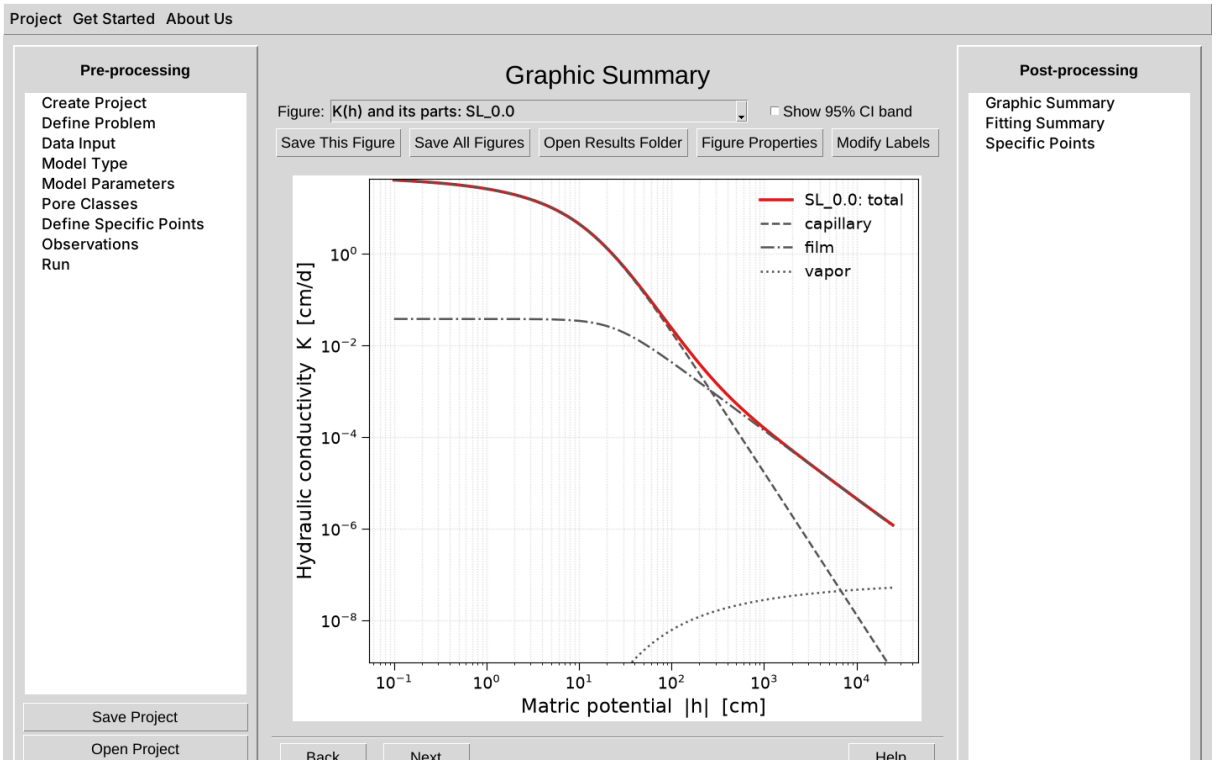


Figure 9.2. Graphic Summary, K(h) and its parts for one sample (PDI van Genuchten, Ks given, vapor included).

The app draws K(h) and K(θ) for all samples, and K(h) split into its parts for each sample. All K values are written to the Excel output in the chosen K unit.

10. Outputs

After a run the app shows three result pages and writes all results to the project folder, in the user's units. Files from an earlier run are replaced, so the folder always matches the latest run.

10.1 Graphic Summary

The Graphic Summary page shows every figure the run produced. A drop-down selects the view; buttons save one or all figures, open the results folder, edit figure properties and modify axis labels. A check box adds the 95% confidence band or error bars.

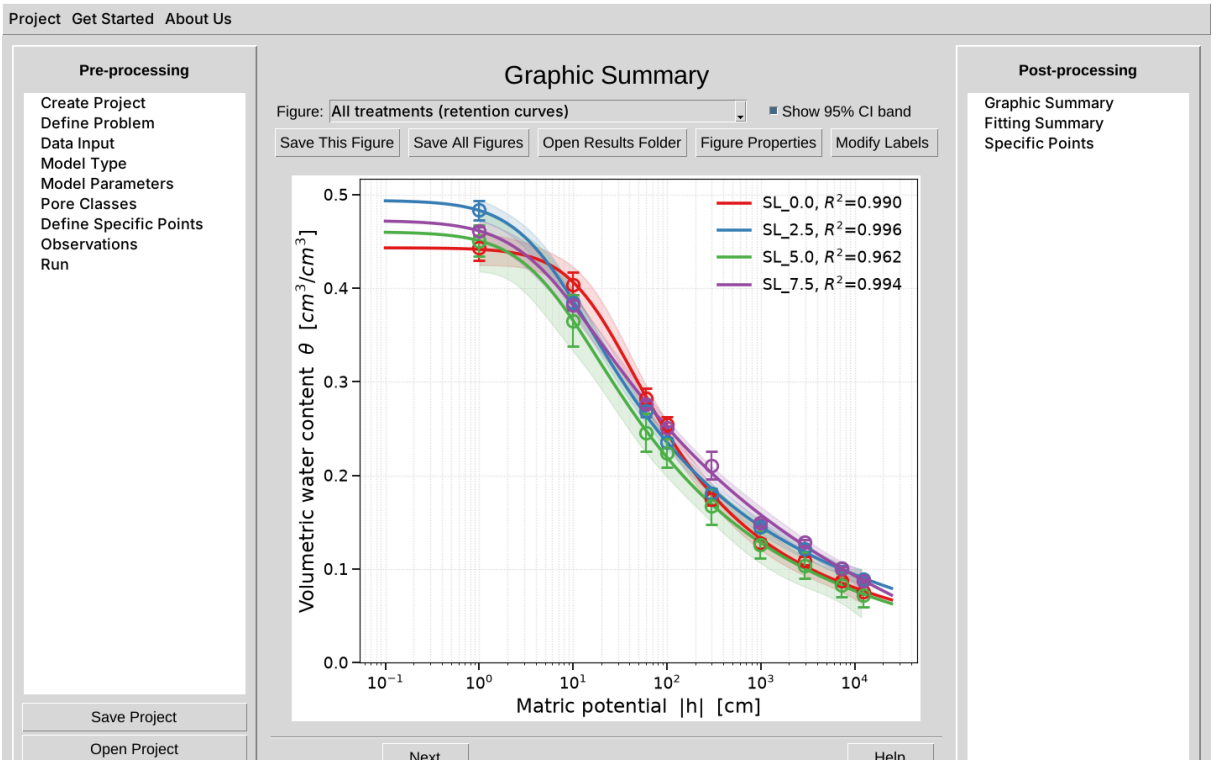


Figure 10.1. Graphic Summary, all treatments with 95% confidence bands. Markers show the mean of replicates with their standard error; the legend gives R² per sample.

View	Content	Shown when
All treatments	Fitted curves and data of all samples	Always
Specific points: water content	θ at FC, PWP and other heads, plus PAWC, as grouped bars	Always
Specific points: matric potential	h at the given water contents, as grouped bars	Points defined
Pore-size distribution	Volume-based $d\theta/d\log_{10} r$ and θ against pore radius	PSD ticked
Pore classes	Class volumes as grouped bars, labels at 45°	PSD ticked
K(h), K(θ)	Conductivity of all samples	K ticked
Treatment	One sample: data, fit and band	Always, one per sample
K(h) and its parts	Capillary, film and vapor parts of one sample	K ticked, one per sample

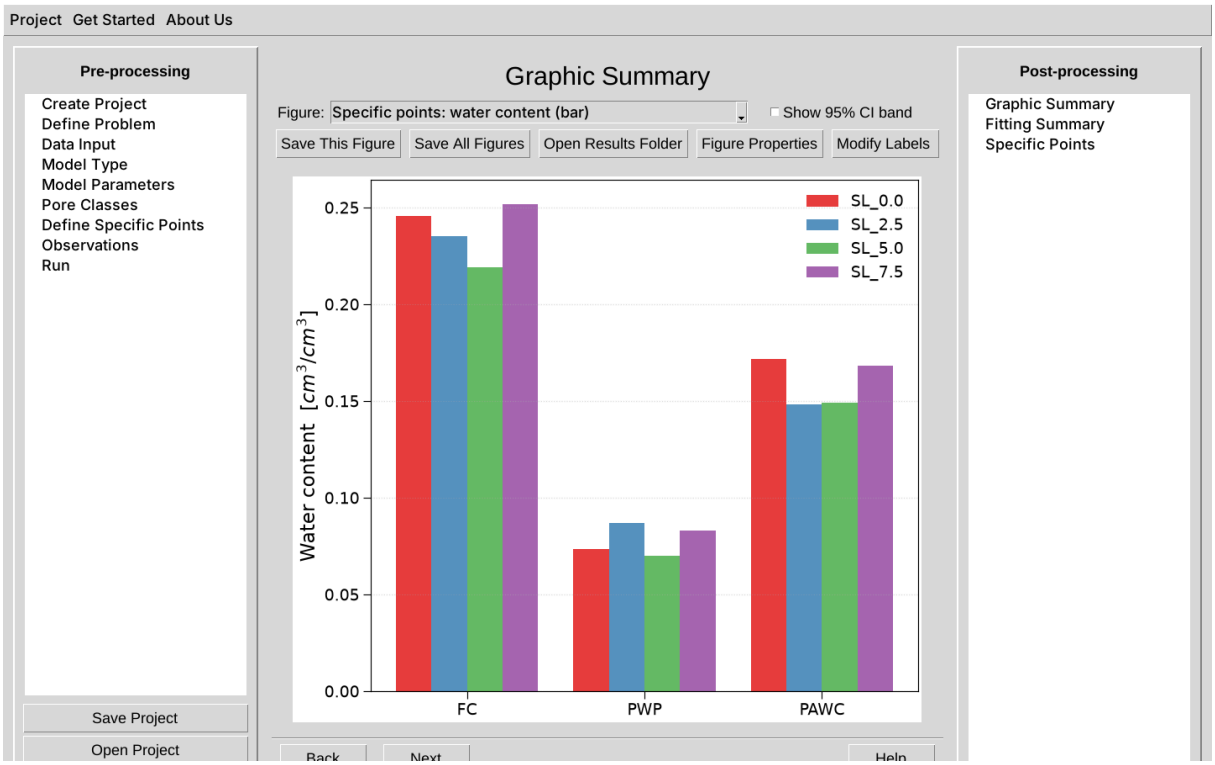


Figure 10.2. Specific points view: θ at FC and PWP and the plant-available water content of the four tutorial samples.

The Figure Properties page sets legend text, marker, color and line style per sample, and font sizes, line widths, grid and resolution for all figures. Settings and edited axis labels are stored in the project file and survive a rerun.

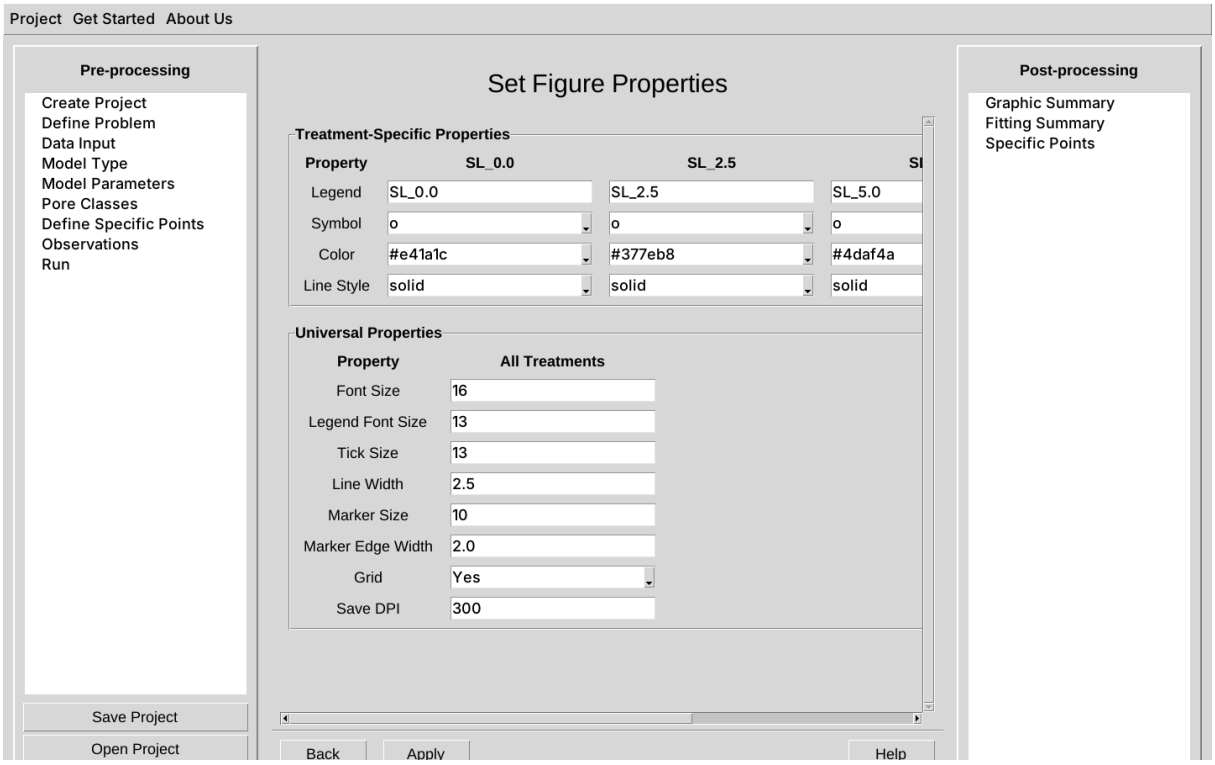


Figure 10.3. Figure Properties page.

10.2 Fitting Summary and Specific Points

The Fitting Summary lists, per sample, the model, each parameter with unit, best value, standard error and 95% interval, notes such as AT BOUND, the parameter correlations, and the goodness-of-fit statistics.

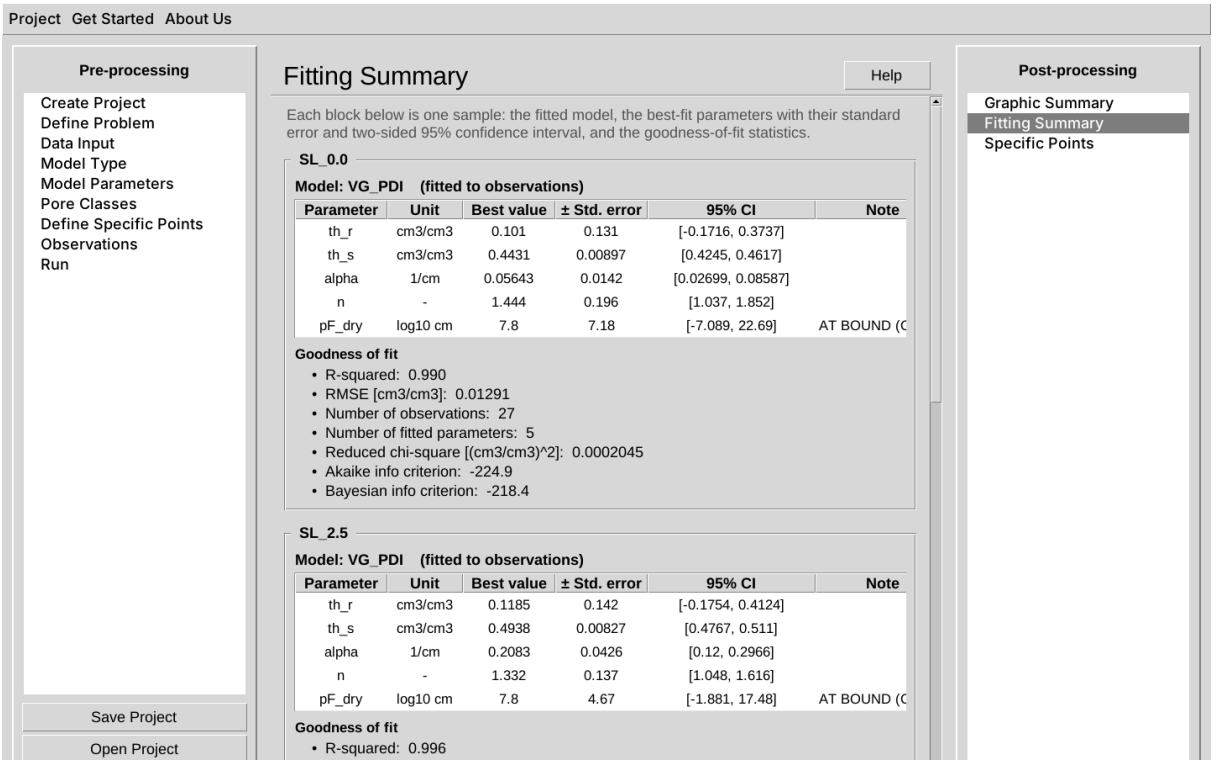


Figure 10.4. Fitting Summary. In this PDI fit pF_dry hits its upper bound (pressure plate data do not reach the dry range), and the app flags it.

The Specific Points page lists θ at the chosen heads, PAWC and h at the chosen water contents, with intervals, followed by the pore-class volumes when requested.

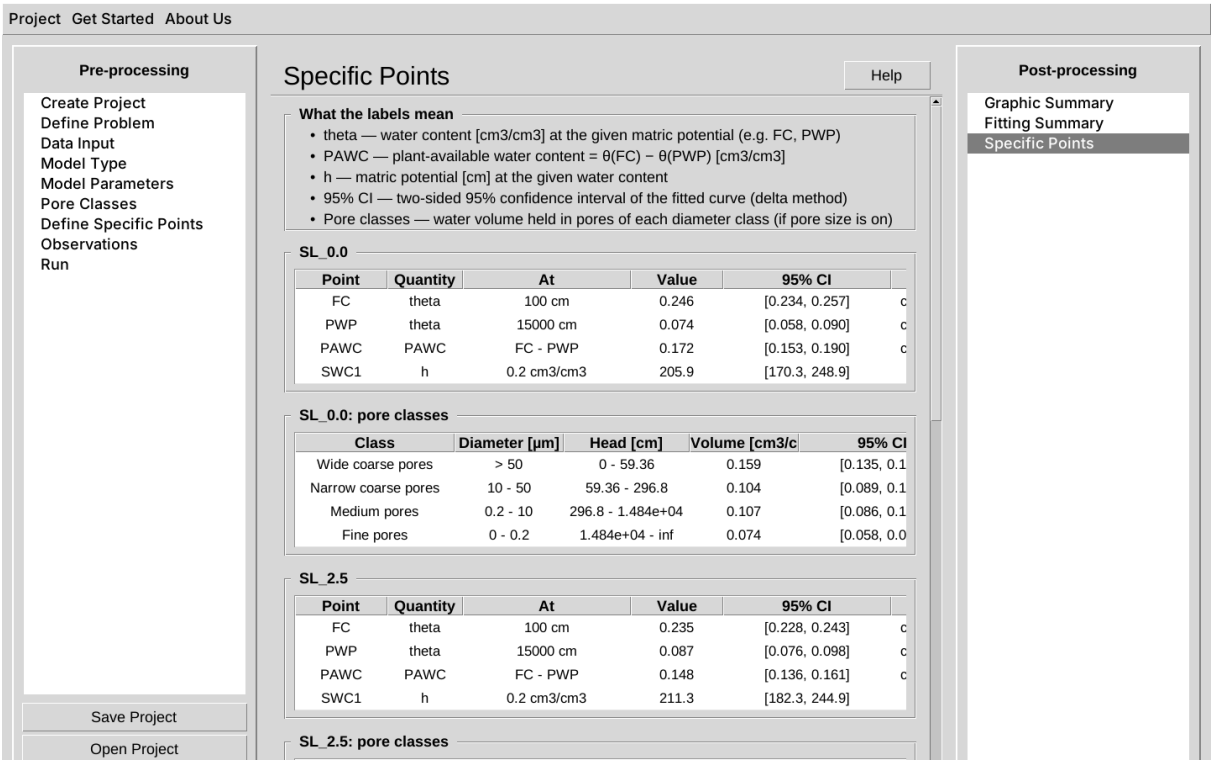


Figure 10.5. Specific Points page.

10.3 Files

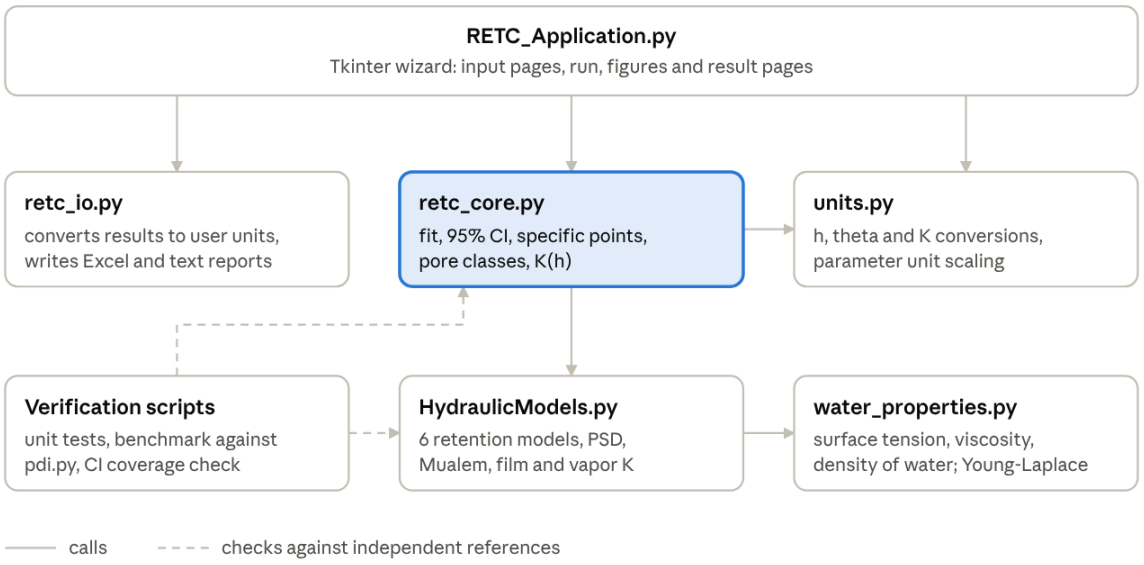
File	Content
Outputs_<sample>.xlsx	Sheets: Retention Curve Data (observed and fitted θ, 95% band, PSD, K columns), ModelParameters, Specific Points, Pore Classes
Summary.xlsx	One row per sample: all parameters with 95% bounds; all specific points; all pore classes
Fittings Parameters.txt	Readable report: parameters, intervals, K settings, correlations, goodness of fit
Specific Points.txt	Readable report: points and pore classes with intervals
Figure_<n>_<view>.png	Every figure, at the chosen resolution (default 300 dpi)
RETC_project.json	All settings, data and figure properties; reopen with Open Project

Units appear in every column header and report line, so files can be shared without the project.

11. Software design and verification

The app is written in Python 3 and split into small modules, each with one task. The science lives in four plain modules with no interface code, so they can be read, tested and reused on their own, for example in the HYPROP post-processing tool that shares this core.

Each module has one task; the interface only calls the science modules



Module	Lines	Role
water_properties.py	57	Temperature functions for σ_w, μ_w, ρ_w; Young–Laplace radius and diameter
units.py	131	All unit conversions and the unit of each parameter
HydraulicModels.py	572	The six models: θ(h), capacity, PSD, Mualem K, film and vapor K
retc_core.py	572	Parameter registry, fitting, statistics, delta-method intervals, specific points, pore classes, K curves
retc_io.py	245	Conversion to user units, Excel files, text reports
RETC_Application.py	3451	Tkinter interface

Design rules:

- **Internal units are fixed** (cm, cm^3/cm^3, cm/d). Conversion happens at the input and once at the output, never inside the science.
- **Parameters are passed by name.** The order of each model's parameters is defined in one registry, so a table row can never be fed to the wrong parameter.
- **One routine per concept.** One delta-method routine serves every interval; one Mualem routine serves every model.
- **Fail loudly.** Negative heads, NaN data, inverted bounds, $n \leq 1$, more parameters than data, all parameters fixed, and singular covariance each stop with a clear message instead of a silent wrong result.

Verification

The science modules are checked by three scripts that ship with the source, plus the coverage test of Section 6.4. All pass for the released version.

Script	What it checks
benchmark_models.py	θ(h) of all six models over pF 0 to 6.8 against independent reference formulas; the PDI models against the published reference code pdi.py of Peters et al. (2024). Largest deviation 2.5×10^{-16}.
test_hydraulic_models.py	Capacity against numerical derivatives; closed-form Mualem integrals against numerical integration (with and without h_clip, both K methods); PSD integrates to 1; Young–Laplace values; PDI dry limit; precision of K in the dry range.
test_retc_core.py	Parameter recovery from synthetic data; delta-method band against lmfit's own band; t quantiles; AT BOUND flags; h(θ) and its interval; weights; guards; singular covariance; unit round trips; K units; pore classes sum to θs; all models fit and export.
coverage_check.py	Monte Carlo coverage of the 95% intervals (Section 6.4).

Requirements and build

The source is tested with Python 3.12 and 3.13 and needs numpy, scipy, pandas, lmfit, matplotlib and openpyxl. A Windows executable is built with PyInstaller using the supplied build_exe.bat and RETC_Application.spec; the program then runs from dist\RETC_Application without a Python installation.

12. Assumptions and limits of use

The results are only as good as the assumptions behind them. The main ones:

- **Rigid, non-hysteretic soil.** All models describe one drainage branch of a rigid pore system. Wetting curves, hysteresis and swelling or shrinking soils are not modeled; for shrinking clays, θs and the pore sizes refer to the state at measurement.
- **Equilibrium data.** Pressure plate points are assumed to be at hydraulic equilibrium. Points that did not reach equilibrium, especially at high suction, bias the dry end of the curve.
- **Extrapolation.** Every derived quantity (PSD tails, fine-pore volume, K near saturation, θ beyond the last measured head) is a property of the model outside the data range. Read values beyond the measured heads as model extrapolations.
- **Capillary bundle.** Pore sizes and K rely on the capillary bundle concept and Young–Laplace with zero contact angle at 20 °C. They are equivalent sizes of the necks that control drainage, not imaged pore bodies.
- **K prediction.** Mualem's model with fixed τ typically predicts K(h) within one order of magnitude. Macropore flow near saturation is not captured; a measured Ks helps only at

saturation. The K band includes only the retention-parameter uncertainty.

- **Linearized intervals.** Confidence intervals assume that the model is approximately linear in its parameters within the interval and that errors are independent and Gaussian. They are not reported when the covariance is singular and are unreliable for parameters AT BOUND. With strong nonlinearity or very few data, profile likelihood or bootstrap methods give more accurate intervals.
- **Model choice.** Bimodal and PDI models need data that cover both pore systems or the dry range. Without such data, fix the extra parameters or use a simpler model.
- **Temperature.** Water properties are evaluated at 20 °C for pore sizes, β and the film and vapor terms.

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